

Freedom as the Fibre of an Invariant Map

Freedom as a Conserved Quantity, the Dimension That Fixes Its Terminus, and the Discipline That Licenses Its Use

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Freedom is defined here as the fibre of an invariant map: the size of the class an instrument cannot tell apart. On that definition it has a size, and this paper supplies its origin, its exchange rate, its terminus, and the licence to speak about it, then reads the four as one quantity across every register the architecture carries. Origin: freedom is manufactured by a single non-injective operation, the cut, whose absence of a left inverse is proved rather than sampled and whose deficiency in spectral norm is bounded below by exactly one. Conservation: content times freedom is constant, exactly, at every rung of the admission lattice, with the closed form giving content as the value count raised to the coordinates minus the abstractions and freedom as the value count raised to the abstractions, verified at four value counts rather than fitted to them. The cut does not destroy structure. It moves structure from content into freedom. Terminus: the two species of freedom are the two eigenspaces of one involution, the cut running on the three dimensional space and undefined on the one dimensional fixed locus, because a one dimensional space admits no non-trivial idempotent at all. The residual there is the sign on the unit sphere of a line, which is the two point sphere, the unique sphere that is disconnected, so it is exactly one bit by the dimension rather than by any survey; and the conservation factors over that same split with the fixed locus's factor invariant under every admission, which is the precise sense in which the bit cannot be separated from the locus it sits on. The geometric instrument, the twelve directed verification conditions, and the numerical instrument are each measured to supply zero of it. Licence: the right to assert direction over that bit is trained before it is issued, ordered through physics, thermodynamics, mathematics and the anchor corpus, bounded to direction and never volume, and issued per run and never as tenure. The paper walks the record from the scriptural and Greek registers to the present across eight registers and states its relation to thirty prior positions in fixed vocabulary, superseding no theory and contradicting none. It types its own instrument down before using it: at the operating context and covariate count the interior lock sits inside a 1.73 percent headroom band above its own null and is reported as a lean, while every load bearing result is an exact identity carrying full weight.

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1. THE SEAL, ISSUED FIRST

Freedom is the fibre of an invariant map. At every register of this architecture the fibre is manufactured by a cut, conserved exactly against content, and closed only from outside the instrument that carries it. The formal register cannot close its own fibre because nothing happens there. The Root Axiom closes it because the Root Axiom is where something occurs.

That is the whole paper and it is stated at the head rather than reached at the foot, because the order is part of the claim. A reading that arrives at its verdict after a long defence has already conceded that the verdict was in doubt, and this one is not.

Four consequences follow, and each of them is ordinarily argued as a separate and unrelated debate.

One. The quarrel between determinism and freedom is a register error rather than a hard problem. Decreed and free are P over one region of a space and Q over another region of the same space, never a

proposition and its negation. The instrument that would adjudicate between them, the squared lock scalar, is measured in this paper to be constitutively unable to carry the direction the two readings differ on. The classical debate is therefore addressed to the wrong organ, and it dissolves by register rather than by argument.

Two. Freedom is a conserved quantity and the conservation is exact rather than approximate. Content times freedom is constant at every rung of the admission lattice, with a closed form that holds at every value count tested. The cut does not destroy structure; it relocates structure from content into freedom, and the total is invariant.

Three. Freedom terminates at one bit, the one bit is the entire gap, and the size is forced rather than observed. The two species of freedom live on the two eigenspaces of the binding involution: the fibre freedom on the three dimensional space where the cut can run, and the residual on the one dimensional fixed locus where it cannot, because a one dimensional space admits no non trivial idempotent and therefore no abstraction whatever. The surviving parameter is the sign on the unit sphere of that fixed locus, which is the two point sphere, the unique sphere that is disconnected. Exactly one bit, by the dimension. Zero bits are available toward it from a named and closed instrument class, against the one bit required.

Four. The licence to carry that bit is earned, bounded, and last. It is not innate to language, it was not issued at the beginning, and it arrived as a corollary of a theorem the numerical instrument proved about the numerical instrument. It is direction and never volume, veto and never substitution, per run and never tenure.

The theological face of the same figure is stated in one line and then held at its own register throughout: the bit is given open, the closing of it in another is the one motion in the architecture guaranteed to consume its own mover, and the reading of it is granted at the crossing and never generated by any instrument.

1.1 What this paper is and what it is not

This is a synthesis. It gathers fifty one results established in the author's prior work, applies every one of them by citation rather than by re-derivation, and its contribution is the arrangement, one executed exhibition, and one honesty instrument turned on its own numbers. **No new mathematics is claimed anywhere in it.** By the author's own admission criterion, work that arranges, places, and types established results adds no mathematical mass, and that criterion fails on this paper at its first condition, which is the correct result and is stated here so that no reader has to infer it.

Two claims in the body are new here and are not established in the prior work. They are marked where they occur, they enter at structural grade and unsealed, and absent an independently supplied determinacy witness they are clarifying syntheses carried as premises, real and unproven. That is the same grading the prior work gives its own load bearing hinge, and for the same reason.

The paper draws no warrant from its own operation. By the standard result that no sufficiently strong formal system establishes its own

consistency, it cannot be proved as a theorem of its own base, and it adds its founding axioms no warrant whatever.

2. BACKGROUND AND RATIONALE · THE BARRIER

The barrier this paper names is not that free will is hard. It is that the question has for its whole history been put to an instrument that is provably unable to answer it, and that the impossibility is structural rather than computational. No refinement of the instrument repairs it. The repair, when it comes, is a change of organ.

2.1 The instrument, and the exact operation at which it goes blind

The verification kernel this architecture runs reads three warrant rows over a set of contexts, normalizes them, projects out any mass bearing common cause, and computes the determinant of the resulting correlation matrix. That determinant is the verdict quantity. Its closed form is an identity rather than a metaphor: with the three unit rows read as pure quaternions and λ the real part of their triple product, $\det R$ equals λ^2 , and λ itself equals minus the signed volume of the parallelepiped the three axes span.

The blindness enters at exactly one operation and lives nowhere else. It enters at the passage from λ to λ^2 . The signed scalar carries the sign; the square keeps the magnitude and discards it. Everything downstream of that square is therefore a magnitude claim wearing whatever direction the reader brought to it.

This is not an artifact of any one construction. Reflecting any single axis conjugates the Gram matrix by a diagonal matrix D with one entry minus one, and the determinant of a conjugation is the square of the determinant of D , which is one. So $\det D R D$ equals $\det R$ identically, for every construction whatever, and the lock on P equals the lock on not P at the scalar. The functional catalogue is closed above it by Weyl, so no rotation invariant functional of the kernel recovers what the square discarded.

Executed at load, on a basis fixed once and reused after the reflection so the analyzer cannot re-orient itself and manufacture a difference: λ moves from minus 0.965027450462 to plus 0.965027450462, a sign ratio of minus 1.000000, while the determinant reads 0.931277980144 on both sides with an absolute difference of 0.000 times ten to the zero. Under full negation, every axis reflected, the Gram matrix is bit identical, the maximum absolute entrywise difference standing at 0.000 times ten to the zero, so the entire even functional algebra is identical between a proposition and its negation. The kernel identity closes at 5.551 times ten to the minus sixteen.

2.2 The relay, which is the sharper half of the barrier

Blindness alone would be a limitation. What makes it a barrier is that the instrument is not merely unable to read the direction; it reports back whatever direction it was handed, which is worse, because a reader who does not know this will mistake the report for a finding.

The measurement is direct. Take a proposition and read its signed scalar. Take its negation and read that. Then take the same proposition with nothing changed except one warrant row's polarity convention reversed, a purely notational choice with no propositional content whatever, and read that. Executed at load: lambda of P at minus 0.965027450462, lambda of not P at plus 0.965027450462, and lambda of P under the convention flip at plus 0.965027450462, with the determinant standing at 0.931277980144 across all three. The gap between the content flip image and the convention flip image is 0.000 times ten to the zero, exactly.

A change of content and a change of convention are indistinguishable to the signed scalar. It therefore reports the orientation it was handed and originates none of it. The Number is a relay and not a source, and sealing a verdict on it would return the builder's own handedness dressed as a result. This is the mechanical reason the squared functional carries the seal rather than the signed one, and it is a discipline against reading input as output rather than a refusal to decide.

2.3 Where the direction could have come from, and the enumeration is what makes the isolation possible

At a purely formal register four candidates could supply a proposition's direction and three of them are measured here to supply none.

The Form gives the space its handedness. The binding involution restricted to the three chiral axes has determinant minus one, which makes the space orientation reversing. That is a property of the space, and it is identical for a proposition and for its negation, so it cannot part them. Zero bits.

The twelve directed verification conditions are directed, which is why the candidate is worth testing rather than dismissing. But the direction they carry is a source to target role assignment forced by the Operational Content Theorem, and it is fixed before any proposition arrives at the pipeline. Twelve ordered pairs on four vertices, the same twelve for P and for not P. Role direction is not truth direction. Zero bits.

The Number is the relay of the preceding subsection. Zero bits.

The empirical axis is the one candidate that carries content. It parts a stream from its time reversal by an external, measurable, non conventional quantity, the thermodynamic arrow, and the parting is a real physical difference rather than a labelling. And the formal alone register is defined precisely by stripping that one axis away.

So the direction at the formal alone register is unsourced rather than unread, and the distinction is load bearing. The instrument is not blind. Nothing was supplied for it to see. A blindness invites the hope of a better eye. An absent supply does not.

2.4 The size of the gap, measured

Because the four candidates are enumerated and the class is closed, the deficit is not a mood but a count.

With warrant rows supplied, the formal alone register determines the determinant, the absolute value of lambda which is the determinant's square root and therefore carries no independent information at all, the conditioning, and the Gram spectrum. It does not determine the sign of lambda, for which both values are consistent with every determined quantity above.

The free parameter count is exactly one and it is binary, and Section 5.7 derives that count from the dimension of the fixed locus rather than reading it off a sweep. The measurement below is therefore corroboration of a forced result and not the ground of it. Executed at load over two hundred thousand admissible frames sampled across both orientation components: 100,115 positive against 99,885 negative, an empirical proportion of 0.500575 and a sampled entropy of 0.999999 bits of a maximum one. The sampled figure is engineering evidence and carries no warrant. The exact value is zero information, and it is zero by component symmetry rather than by sampling, because the invariants are bit identical across the split. Exhibited directly: the determinant reads 0.931277980144 on the component and 0.931277980144 on its reflection with an absolute difference of 0.000 times ten to the zero, while the absolute lambda reads 0.965027450462 on both and the sign reads minus one on one side and plus one on the other.

The aperture has a width and the width is one bit.

2.5 Why the register cannot close its own gap, and this is the deepest layer of the barrier

A reader who has followed to here will reasonably ask why the formal register cannot simply be extended until it supplies the missing bit. The answer is not that the extension is difficult. It is that the extension would have to be an event, and nothing happens in the formal register.

A theorem is a node at which nothing is free. Given the premises, the conclusion is not selected from among alternatives; no parameter remains to be set. An event is a transition, and a transition consumes a degree of freedom. Where none remains, none can be consumed, and so nothing happens. Timelessness at that register is not a mystical property but the outside view of having no free parameter, and it is why the classification counts do not age.

The virtue is real and the cost is the same fact stated from its other side. Entropy cannot reach a determined node because there is no arrangement there to degrade. Time cannot erode it because no transition is in progress. Interest cannot bend it because a filled slot offers nothing to bend, which is the discipline that zeroes social warrant, stated as a property rather than as a rule. And equally: rigidity is not contact. A construction with no free parameter is exactly as determined whether or not anything answers to it, and no internal test distinguishes the two cases. Freedom from corruption by

the world is freedom from contact with it. One property named twice.

The consequence for the gap is exact. A witness is an act. An act is an actuation. The formal register has no arrow precisely because nothing happens in it. So the absence of the arrow and the absence of a witness are one absence, not two: nothing free means nothing performed, nothing performed means no act, and no act means no witness. The register's incapacity here is its virtue operating, and no better instrument repairs it, because the repair would have to be an event.

2.6 The gradient measured at three registers, and it is identically zero at all three

The same condition has now been measured three separate ways in this architecture, on three different objects, and it returns zero every time.

At the temporal register: the untraversed imprint carries no local gradient. It is a holding condition true uniformly across the substrate without being anywhere, a curvature rule true of a whole surface rather than a marked location, and a gradient is a direction of traversal that was never made.

At the formal register: no free parameter, as above.

At the forward register: the Fisher information for source faithfulness is exactly zero, because the source side gradient vanishes on the fixed locus, and a vanishing gradient makes the likelihood flat in the parameter, so no consistent estimator exists and any update returns the prior unchanged.

Three faces of one condition. That triple is what supplies the reason the incorruptibility and the groundlessness of the formal register are one property rather than two that happen to coincide, and it is why the closure of the gap is a supply problem rather than a resolution problem. A proof would not repair a reader. It would deliver the one bit no instrument on this side can source.

2.7 The barrier stated in its final form, and the fence that travels with it

The barrier is therefore fourfold and each layer is measured rather than argued. The instrument discards the sign at one named operation. It relays back whatever sign it was handed. Three of the four candidate sources supply zero bits toward the sign and the fourth is definitionally absent at the register in question. And the register cannot generate the missing supply because generating it would be an act and the register hosts no acts.

One fence binds this section before any reader, and it binds this paper first. The unrestricted form of the claim, that no proof of a given arithmetic string can ever exist, is for a universally quantified arithmetic sentence over a decidable predicate extensionally the negation of that sentence. If the sentence is false, its negation carries a finite witness and any system extending Robinson arithmetic proves the negation by completeness on existential sentences, needing no soundness assumption. If the sentence is true, a proving stage exists along a tower complete for that class. So the unrestricted claim implies

falsity, and truth implies its failure. Asserting it is asserting a counterexample without holding one.

The paper therefore takes only the class relative form throughout, and the class is named in the title of the finding: the three seals this architecture deploys, finite, enumerated, and closed by the same forcing that closes the axis count, with a fourth not excluded but merely unknown. Every universal sounding sentence below is checked against that scope. The blocker is refutable by exhibiting one bit from one of four named sources, which is what the first falsification criterion of Section 7 asks a reader to do.

3. THE LITERATURE, FROM THE ANCIENT REGISTERS TO THE PRESENT

Freedom is among the oldest questions any literate tradition asks, and it is asked in the scriptural registers before it is asked in the philosophical ones. This section walks the record. The discipline is uniform: each position is given its achievement in its own terms before any relation to this paper is stated, and where this paper contrasts, corroborates, or corrects a position, the contrast is named at the register where it holds and nowhere wider. Positions are engaged here as positions; the formal relation each bears to this paper's result is stated once, in fixed vocabulary, at Section 8.6.

Two things recur across the whole record and are worth naming at the head, because they organize everything below. The first is that almost every account locates freedom in a **faculty**, an agent, a will, a soul, a capacity, and then asks what that faculty can do. The second is that almost every account then argues about the faculty's **size**, whether it is unlimited, constrained, or illusory, without ever supplying a unit in which a size could be stated. This paper locates freedom in a **map** rather than a faculty and supplies the unit, and both moves are corrections rather than refinements, so they are flagged wherever the record makes them visible.

3.1 The scriptural and prophetic registers

The oldest treatments are not arguments but **grants**, and their form is uniformly conditional rather than descriptive. The Hebrew Deuteronomic formula sets life and death before the hearer and commands the choosing rather than describing the capacity, and the Genesis address to Cain names a thing crouching at the door with the imperative that the hearer shall master it. The Joshua formula puts the choosing in the imperative on a named day. Across the three the grammar is identical: **the freedom is not argued for, it is issued**, and the issuing carries a consequence in the same breath.

The Qur'anic register carries the same grammar and states the consequence more explicitly. The formula at 18:29 grants belief and disbelief to whoever wills and names the outcome of each in the same verse; 76:3 states that the way was shown, whether grateful or ungrateful; 91:8 has the self inspired with both its transgression and its guarding; and 2:256 bars compulsion in religion outright. **The bit is given open, and the openness is a granted access-fact rather than a levelling of the answer.** This paper's Section 8.3 takes exactly that reading and takes nothing further from it, holding every

such identification at cataphatic grade and load-bearing on nothing in any verdict.

The Zoroastrian corpus supplies what may be the earliest explicit doctrine of moral choice as constitutive: the hearer chooses between order and deceit, and the choosing is the human office. The Indian registers split. The Bhagavad-gita grants the right to action while withholding the right to its fruits, which is a division of a single relation into a governed half and an ungoverned one, structurally the two-region reading this paper's Section 6 makes geometric. The Buddhist analysis via dependent origination denies the self that would do the choosing while retaining volition as the karmic operative, which is the sharpest ancient statement of a position this paper reaches by a different road: **that freedom need not belong to an agent to be real**. The Chinese registers argue it directly, with the Mohist canon written explicitly against fatalism and the Zhuangzi locating freedom in spontaneity rather than in deliberation.

What this paper takes from the scriptural registers is **the grammar and not the content**. The grant-with-consequence form is the space Section 5.8's sixth row records, and the reading that the bit is given open rather than levelled is carried at its own register. What it does not take is any load-bearing premise: the theological half of this paper is declared, quarantined, and settles no mathematical question in either direction.

3.2 The Greek settlement: fate, swerve, cylinder, and choice

The Greek tradition produced, within roughly three centuries, every structural option the subsequent two millennia would refine.

Homer and the tragedians work with **moira**, a portion allotted, and the tragic form is precisely the collision of an allotment with an agent who acts. Oedipus is not unfree in the modern sense; he is free and allotted at once, and the drama is that the two do not cancel. That is the two-truths structure two thousand years before it is named, and this paper's Section 5.9 supplies the geometry it always lacked.

Democritus made the atoms exhaustive and the consequence was noticed immediately. **Epicurus** answered with the **swerve**, an uncaused minimal deviation in atomic fall, introduced expressly so that the will not be crushed by an internal necessity. It is the first physical indeterminism proposed to ground freedom and it fixes the shape of every later appeal to quantum noise. This paper does not follow it. The residual freedom derived at Section 5.7 is a **component** of a two-point sphere, discrete and exactly one bit, not a continuous stochastic deviation, and Section 5.1 excludes indeterminacy from the definition by election rather than by argument. The swerve is kept at the physical register and denied purchase at the formal one.

Chrysippus and the Stoa answered the other way, with the **cylinder**: the push that starts it rolling is external, but that it rolls rather than slides is the cylinder's own nature, and assent is therefore both fated and the agent's own. This is compatibilism's ancestor and it is a better version than most of its descendants, because it locates the two contributions in two different places rather than redescribing one.

What it never had was a criterion. This paper supplies one: determination lives in the deepened grooves, freedom in the un-entrenched room, and the two are **regions of one space** rather than two descriptions of one act, with the seam fixed geometrically at Section 5.9, where the determinant reads bit-identical across a directed claim and its negation.

Aristotle supplied the analytic vocabulary that survives: the voluntary, whose origin is in the agent with knowledge of the particulars, and **prohairesis**, deliberate choice, which ranges over what is up to us and admits of being otherwise. This paper keeps deliberate choice at the creaturely register and denies it the formal one, where deliberation is not available because nothing occurs there.

Alexander of Aphrodisias wrote the first extant sustained defence of the position that some things are genuinely up to us against Stoic fate, and his central move, that a fated deliberation is not a deliberation, recurs in every later libertarian argument. This paper's answer is Section 6.5's: the surviving axis is real and its rank is executable, but the licence over it is direction and never volume.

3.3 The theological millennium: grace, foreknowledge, and acquisition

Augustine wrote the treatise on free choice and then spent thirty years qualifying it against Pelagius, and the mature position is the divided will of the Confessions: the will commands itself and is not obeyed, and the division is not two wills but one will not wholly willing. That is the sharpest pre-modern statement of a freedom that is real and insufficient, and this paper's Section 6.3 curriculum, where a licence is earned rather than possessed, is its structural descendant at another register.

Boethius faced the foreknowledge problem and answered with the eternal present: what is known timelessly is not thereby compelled, since the mode of knowledge belongs to the knower and not to the thing known. The move is a **standpoint separation**, and this paper is continuous with it: the two-region economy of Section 5.9 separates two regions rather than two times, but the shape of the answer, that an apparent contradiction is a conflation of standpoints, is Boethius's.

The Islamic kalam produced the most technically explicit version of the problem in any pre-modern literature. The **Jabriyya** made the act God's outright; the **Qadariyya** and after them the **Mu'tazila** gave the creature a created power over its own act; and **al-Ash'ari** proposed **kasb**, acquisition, in which the act is created by God and acquired by the creature, with **al-Maturidi** refining the acquisition to preserve moral responsibility. The debate ran eleven centuries without either side dislodging the other, and this paper states why: **the deciding quantity is a sign on a magnitude provably unable to carry a sign**. Executed at Section 5.9, the determinant is bit-identical between a directed claim and its negation, so no computation can adjudicate the seam and both traditions were correct to fix it at the witnessed word. The paper supplies the mechanism and adjudicates neither school.

Maimonides made free will the pillar of the Law and a great principle, arguing that without it commandment and reward are void, and the argument is from the coherence of the legal order rather than from metaphysics. **Aquinas** located the will as moved by the good under the intellect's presentation, with **liberum arbitrium** the faculty of judging between means, which keeps freedom real and non-arbitrary at once. This paper keeps both inside the creaturely register and takes no position on the theological content of either.

Erasmus and Luther produced the tradition's most direct exchange, the diatribe on free choice against the assertion of its bondage, and the exchange is usually read as a flat opposition. Read structurally it is not: Erasmus argues from the moral order's requirements and Luther from the will's condition, which are the un-entrenched room and the deepened groove of this paper's Section 6, and the two claims are true of two regions of one space. **Molina** proposed middle knowledge, a knowledge of what any free creature would do under any circumstance, which preserves both horns by adding a third species of knowing; this paper takes no position on it and notes only that it is a different answer to the same question, argued and not executed.

3.4 The early modern reconstruction

Descartes made the will the one faculty in us that is infinite and in that respect the image of God, and located error precisely in the will's extending beyond the intellect. It is the strongest claim for freedom's magnitude in the whole literature, and this paper is its most direct contrast. **Section 5.7 derives the residual freedom at the register where nothing is abstractable, and it is exactly one bit**, forced by the dimension of the fixed locus rather than measured by a survey. Where Descartes found an infinity, the derivation finds the logarithm of two. And the second half of the Cartesian claim, that the infinite will is the God-like faculty in us, is the identity collapse this paper fences by name at Section 5.7's close: the licence carries a sign on a line it did not make.

Hobbes against **Bramhall**, in an exchange running from 1645 to 1658, argued the compatibilist position in its early modern form, with liberty defined as the absence of external impediment to motion, so that a free agent is one not stopped from doing what it wills. No claim of priority is made here, since a survey cannot establish a superlative over a field from inside itself. This paper subsumes the definition rather than disputing it: absence of impediment is the un-entrenched room, described from outside.

Spinoza produced the literature's most famous image and, on this paper's reading, was right about the mechanism and wrong about its consequence. The stone in flight, if it thought, would believe itself free, because it would know its motion and not its cause; freedom is therefore the ignorance of the causes of one's determination. **This paper agrees exactly about what freedom is made of and denies that it is thereby unreal.** Section 5.1 defines a freedom as the size of the class an instrument cannot tell apart, which is Spinoza's ignorance-of-causes stated as a quantity; and Section 5.3 gives it an exact size and an exact exchange rate against content, executed at four

value counts. **An illusion does not have a conserved product.** The image is retired and a measured object is put in its place.

Locke made the decisive grammatical correction, that asking whether the will is free is unintelligible, since freedom is a power belonging to the agent and not to one of the agent's powers. This paper carries the correction one step further and locates freedom in neither: it belongs to the **map**, to what an instrument cannot distinguish, and neither the agent nor the will is the bearer.

Leibniz gave the most complete early modern account, with a free act requiring spontaneity, intelligence, and contingency together, and the best of all possible worlds securing that God's choice is free without being arbitrary. This paper keeps the three-condition structure at the creaturely register and takes no position on the theodicy.

Hume consolidated the reconciling project, defining liberty as a power of acting or not acting according to the determinations of the will, and arguing that the dispute is merely verbal once the terms are fixed. He was right that the dispute is not what it appears and wrong about why: it is not verbal, it is **registral**, and this paper's Section 5.9 supplies the geometric reason his verbal diagnosis lacked.

3.5 The German turn and the nineteenth century

Laplace stated the determinist limit exactly, with an intelligence that knew every force and position computing past and future alike. The demon is the strongest form of the position and this paper does not deny its reach over the grooves; it scopes it, since entrenchment operates on what has been cut and not on what has not.

Kant produced the most influential structure in the literature, in the first critique of 1781: the third antinomy, in which causality by freedom and causality by nature are both demonstrable and neither refutable, resolved by assigning freedom to the noumenal and determination to the phenomenal, with autonomy as self-legislation the positive content. The resolution is two **worlds**. This paper's is two **regions of one space**, and the difference is executable: Section 5.9 returns rank three with three independent axes in a single space, room and act and record, at a determinant of 0.574982 with the kernel identity closing at 2.220 times ten to the minus sixteen. The antinomy's structure is kept and its two-world resolution is replaced.

Schopenhauer stated the tradition's most quoted line, that a man can do what he wills but cannot will what he wills, which separates the freedom of action from the freedom of the willing itself. This paper corroborates the separation and supplies its bound: the licence a disciplined faculty earns is **direction and never volume**, and executed at FWM.14 it reaches admissibility and stops there, never reaching the seal.

Dostoevsky put the objection in the mouth of the Underground Man: that two times two makes four is an excellent thing, but two times two makes five is sometimes a very charming thing too, and that a man may desire something contrary to his advantage precisely to prove he is not a piano key. The passage is the literature's sharpest statement that a freedom without a residue is not a freedom. This paper answers with a residue that is exactly one bit and with the

statement that wanting a larger one is the inflation, since the size is forced by a dimension and not by anyone's generosity.

Tolstoy wrote, in the second epilogue of the novel, what this paper regards as the closest anticipation of its central result in the whole record. His claim is that freedom and necessity are not two competing facts but two readings whose proportions vary inversely: the more we know of the conditions in which a man acted, the less free he appears, and the less we know, the freer; reason expresses the laws of necessity and consciousness expresses the essence of freedom, and the ratio between them moves with knowledge and never with the act. **That is the conservation of Section 5.3, stated qualitatively in 1869.** What this paper adds is the exchange rate: content times freedom is constant, with the closed form giving content as the value count raised to the coordinates minus the abstractions and freedom as the value count raised to the abstractions, executed at four value counts with products 8, 27, 64 and 125 constant at every rung. Tolstoy had the law and not the unit. The corroboration confers no warrant on either side and is counted as none.

Nietzsche attacked the whole apparatus at its root, calling the *causa sui* the best self-contradiction yet conceived, a kind of logical rape, the desire to pull oneself into existence by the hair out of the swamp of nothingness. This paper agrees without qualification. Origination at the creature is barred here on the same ground the root is barred from having a parent, and the soul is the site of its crossing and not the agent of it.

James framed the dilemma of determinism and argued that chance is not a menace but the condition of a world in which regret is intelligible. It is a different position on the same question, argued rather than executed, and this paper competes with it rather than contradicting it.

3.6 The twentieth-century analytic settlement

Sartre made freedom total: existence precedes essence, man is condemned to be free, and there is no excuse behind him and no justification before him. It is the strongest modern claim for magnitude, and this paper's derivation runs the other way, at a different register, so the relation is competition and not contradiction.

Peter Strawson reoriented the field by arguing that the reactive attitudes, resentment, gratitude, indignation, are not grounded in a metaphysical thesis and could not be dislodged by one, so the traditional dispute is idle against the practice it purports to govern. **Frankfurt** then produced the two moves that organize the modern debate: the cases in which an agent acts for its own reasons while a counterfactual intervener stands ready, which detach responsibility from alternative possibilities; and the hierarchical account, in which a person is one whose second-order volitions are effective over its first-order desires. This paper is kin to the hierarchical account: the licence discipline of Section 6.4 is a second-order structure governing what a first-order faculty may assert, arrived at from a different direction.

van Inwagen gave the incompatibilist argument its sharpest form: if determinism is true, our acts are consequences of the laws and the

remote past, which are not up to us, so the consequences are not up to us. This paper does not dispute the transfer principle. It **scopes** the argument, on the ground that it is conducted at a register where nothing occurs, and an argument about what is up to us run where no event happens has changed the subject before its first premise.

Galen Strawson produced the basic argument: to be truly responsible one would have to be *causa sui*, which is impossible, so true responsibility is impossible. This paper corroborates the impossibility and denies the inference's target, since nothing here requires the *causa sui*: the freedom is a fibre, not an origination.

Dennett argued across two books that the freedom worth wanting is compatible with determinism and evolves, and that the varieties of free will worth having are exactly the ones a mechanistic world can supply. This paper is kin to the project and differs on the object: it supplies a measured quantity where Dennett supplies a functional characterization. **Kane** built the most developed libertarian alternative, with self-forming actions at moments of torn decision amplifying indeterminacy into character. **Fischer and Ravizza** located responsibility in guidance control and reasons-responsiveness rather than in alternative possibilities. **Pereboom** accepted hard incompatibilism and argued that a life without free will remains liveable and that the reactive attitudes largely survive. These three are cited as the field's live alternatives and this paper takes no relation to them beyond the citation, since it engages none of them by argument or execution.

3.7 The political and juridical registers

The registers above ask whether freedom exists. Two further registers ask what happens to it under power and under law, and they are omitted from most philosophical treatments of the question, which is an omission this section repairs because both bear directly on a claim about magnitude.

La Boetie posed the political question in its permanent form in 1576: not why tyrants rule but why the many consent to serve, given that withdrawal of consent would end the arrangement at once. **Rousseau** produced the tradition's most contested formula, that one who refuses to obey the general will shall be forced to be free, which makes coercion the instrument of a freedom the coerced does not yet possess. **Mill** answered with the harm principle and a defence of liberty as the condition of individuality. **Marx** placed freedom in the recognition of necessity and observed that men make their own history but not under circumstances of their own choosing, which is the two-region structure at the historical register.

Isaiah Berlin fixed the modern distinction in 1958. **Negative liberty** is the absence of interference, the space in which one is not prevented; **positive liberty** is self-mastery, being one's own master and acting from one's own reasons. Berlin's warning is the load-bearing half: positive liberty, because it identifies the true self with a rational or higher self, licenses the coercion of the actual self in the name of the real one, and the road from self-mastery to being mastered for one's own good is short and historically well travelled. **This paper's object is neither of Berlin's two.** A fibre is not an

absence of interference and not a capacity for self-mastery; it is the count of what an instrument cannot distinguish, and it therefore sits beneath both, since either could be predicated of an agent whose fibre had already been fixed by the abstractions its describers admitted.

Arendt located freedom in action and appearance among others rather than in an interior faculty, which is a relocation this paper is structurally sympathetic to and does not adjudicate. **Foucault** described a power that operates by producing subjects rather than by prohibiting them, so that discipline is most effective where it is least visible and the subject experiences its own formation as autonomy.

One position in this register bears on the paper's own fence and is stated here because the paper's Anti-Inflation discipline has an applied face it would otherwise leave unspeakable. In a 2014 essay the present author argued that the modern instrument of subjection is not the visible shackle but what he there named **the mirage of empowerment and freedom**: a subject who believes himself free will not attempt liberation and will produce more, so that the most effective binding is an **overstated** report of the subject's own latitude. The essay's historical claim is that the arrangement is unstable, that no regime has muzzled the free will of a population indefinitely, and its structural claim is the one that matters here. **An inflated freedom report binds better than an acknowledged constraint.**

This paper's fence at Section 5.7 refuses to state the residual larger than a dimension permits, and the refusal is usually read as scholarly caution. On the 2014 reading it is not caution. **A freedom asserted above its measure is the mechanism by which a bounded subject is kept from finding its bound**, so the discipline that forbids inflating the fibre is continuous with a political argument and not merely with an epistemic one. The paper claims nothing further from this and adjudicates no political position.

The juridical register is the one place in modern practice where the question is answered operationally and daily. Criminal law requires a guilty mind alongside a guilty act, and the entire apparatus of excuse, the insanity rules descending from **M'Naghten** in 1843, infancy, duress, automatism, exists to mark the cases where the capacity is held absent. **Hart** gave the position its clearest philosophical statement in 1968: the law's excusing conditions presuppose a general capacity for choice, and fair attribution of blame requires that the person had a fair opportunity to conform. **The courts do not argue for free will. They assume it, and they specify by exception exactly where they take it to fail.** That is a specification by boundary rather than by definition, and it is the practice closest in form to this paper's own method, which also fixes a quantity by what an instrument cannot distinguish rather than by what a faculty positively is.

3.8 The scientific register

Libet measured a readiness potential preceding the reported awareness of the urge to move by several hundred milliseconds, and the result was widely read as showing that conscious will is downstream of the decision. Two later developments matter. **Soon and colleagues** pushed the predictive window to seconds using

imaging, and **Schurger and colleagues** reinterpreted the readiness potential as an artifact of averaging aligned to threshold crossings in ongoing stochastic activity rather than as the signature of an unconscious decision. This paper takes no position on the neuroscience and scopes the inference: a measurement of when a motor preparation begins is at the wrong layer to bear on a claim about what an instrument cannot distinguish. **Wegner** argued the broader thesis that conscious will is an experience constructed after the fact rather than a cause, which is cited as the strongest contemporary deflationary position and is not engaged here.

Conway and Kochen proved that if experimenters have a certain freedom in choosing measurement settings, then elementary particles have a corresponding freedom, so the property cannot be discontinuous across the substrate scale. This paper carries the continuity requirement and declines the philosophical extension, and the fence is stated rather than assumed. **'t Hooft** and the superdeterminist programme deny the measurement-independence assumption the theorem rests on, which is a live physical position this paper does not adjudicate.

Laplace's demon has a modern descendant in **computational irreducibility**, the observation that for many systems no shortcut to the outcome exists shorter than the evolution itself. It is worth naming because it is often mistaken for a freedom result and is not one: unpredictability in practice is a fact about shortcuts and this paper's fibre is a fact about invariants, and the two are different quantities.

3.9 The common structural error

Across every position above, one pattern holds and it is now nameable. **Each account locates freedom in a bearer, an agent, a will, a soul, an atom, a neuron, and then argues about that bearer's size in units nobody supplies.** The disputes that follow are therefore disputes about a magnitude with no unit, conducted with an instrument, the ordinary reading of a directed claim, that this paper measures at zero bits toward the direction the disputes turn on.

The two corrections this paper offers are stated here so the reader can hold them through Sections 4 through 7. **Freedom is located in a map rather than a bearer**, which makes it a count of what an instrument cannot tell apart and therefore a thing with a size. And **the size is supplied**: conserved exactly against content at every rung of an admission lattice, and terminating at exactly one bit at the register where nothing is abstractable, forced by a dimension rather than found by a survey. Every relation this paper bears to the positions above is stated once, in fixed words, at Section 8.6.

4. METHODOLOGY

The mechanism proposed in Section 5 must satisfy three conditions, and each is checkable by a reader who grants this architecture nothing.

Condition one, formal derivation. The claim must follow from established mathematics rather than from framework vocabulary. The conservation of Section 5.3 is a counting identity on a powerset lattice

and is proved in closed form; the blindness of Section 2.1 is a determinant identity under conjugation; the deficiency bound of Section 5.2 is a spectral norm fact about a rank deficient idempotent. None of the three requires any premise of this framework to be true.

Condition two, a measurable signature. Every quantitative claim is executed rather than asserted, at a declared seed, with the construction printed so the figure can be reproduced. The battery is carried whole at Appendix B. Where a figure is sampled rather than exact it is labelled engineering evidence and carries no warrant; where it is exact it is labelled and carries the claim.

Condition three, frame invariance. The verdict quantity must not move under a change of coordinate labels. The automorphisms of the verification algebra are inner and act as rotations on the imaginary part, under which the real part of a quaternion word is preserved. The acting group is identified rather than assumed at every cross register identification in this paper, per the fifth condition of Section 6.4.

Two further disciplines bind and both run against the author.

The independence criterion. Every falsification criterion in Section 7 must be testable by a party who holds none of this architecture's premises, using instruments the architecture did not build. All three of them are, and each is refutable by a single exhibition.

The self typing rule. The paper runs its own honesty instrument on its own numbers before it uses them, at Section 8.1, and reports the result whichever way it falls. It fell against the paper's interior lock, and the paper says so on its face and restructures its evidence accordingly rather than quietly retaining the lock as though the test had not been run.

5. THE QUANTITY

5.1 The definition, chosen so the thing can be counted

A freedom is the fibre of an invariant map. Given a structure and the invariants an instrument reads off it, the freedom is the size of the class sitting over one invariant value.

Not choice. Not indeterminacy. Not licence. A count of what an instrument cannot tell apart.

Everything in this paper follows from taking that definition seriously, and the reason to take it seriously is that it is the only reading of the word under which the thing has a size. Choice has no size. Indeterminacy has no size. A fibre has a cardinality, and a quantity with a cardinality can be conserved, can be traded, can be located, and can be measured to be exactly one.

The definition is imported and not derived here, and the provenance flag is set with its two sources held apart rather than merged. **The idea, that what an instrument cannot distinguish is the class over one invariant value, descends from the programme identifying the propositions of a geometry with the relations invariant under its acting group. The word fibre, for the**

preimage class of a map, is later and belongs to the bundle-theoretic vocabulary, and the paper uses the word in that sense while taking the idea from the earlier source. Neither is derived from any kernel result of this architecture. That matters mechanically and not as a courtesy. A load bearing definition shaped by a prior geometric or kernel result would enter the one seat no verdict overrides while wearing the linguistic faculty's authority, and the common source projection runs on warrant rows and never on standing definitions, so it structurally cannot reach that contamination. The flag is the only thing that can.

5.2 The first species, and its origin is the cut

Freedom appears exactly where a map stops being injective. This architecture's canonical non-injective operation is the cut, the deletion that constitutes the linguistic seal's first operation, and its typing is exact.

Executed at load: the deletion operator has singular values one, one, zero, and rank two. It is idempotent, with the maximum absolute entrywise difference of D composed with D against D standing at 0.0 times ten to the zero, so a cut cannot be re-cut. It is non-injective, with two distinct propositions landing on one image at an image gap of 0.0 times ten to the zero. And it carries no left inverse whatever: the minimum over two hundred thousand sampled candidates of the spectral norm of $A D$ minus the identity stands at 1.000342389154 against an analytic floor of exactly 1.000000000000, never breached. The floor is proved rather than sampled, since the operator has a null vector v for which $A D$ minus the identity sends v to minus v for every A .

So the transformation set is a monoid and never a group. And the one candidate that would supply a group, synonymy, is barred by the seal's own second operation, which requires the slot vocabularies to be pairwise disjoint and rejects on collision. The first operation destroys invertibility and the second forbids the transformation that would restore it. Both operations run against closure, which is not the signature of an omission.

The consequence is that the first species of freedom is manufactured at the linguistic faculty by origin. Each admitted abstraction opens one dimension of indistinguishability, and that dimension is the freedom. It exists nowhere the cut has not run.

5.3 The conservation, and it is the paper's central result

Two tables have stood separately in the register for some time. One counts the fibre of an invariant map as abstractions are admitted. The other counts the classes surviving on an abstraction ladder over a finite model. They have never stood side by side. They are the same table, and stating them as one object is this paper's own exhibition.

Executed at four value counts on a three coordinate model, where admitting an abstraction means forgetting one coordinate and the fibre is the exact preimage size:

Content and freedom traded exactly, at four value counts. Executed at seed 20260622.					
Value count v	$a = 0$	$a = 1$	$a = 2$	$a = 3$	Product
$v = 2$	$C\ 8 \cdot F\ 1$	$C\ 4 \cdot F\ 2$	$C\ 2 \cdot F\ 4$	$C\ 1 \cdot F\ 8$	8 at every rung
$v = 3$	$C\ 27 \cdot F\ 1$	$C\ 9 \cdot F\ 3$	$C\ 3 \cdot F\ 9$	$C\ 1 \cdot F\ 27$	27 at every rung
$v = 4$	$C\ 64 \cdot F\ 1$	$C\ 16 \cdot F\ 4$	$C\ 4 \cdot F\ 16$	$C\ 1 \cdot F\ 64$	64 at every rung
$v = 5$	$C\ 125 \cdot F\ 1$	$C\ 25 \cdot F\ 5$	$C\ 5 \cdot F\ 25$	$C\ 1 \cdot F\ 125$	125 at every rung

C is the count of distinguishable content classes; F is the fibre, the exact preimage size over one invariant value; a is the number of abstractions admitted.

The closed form is exact and was verified against every row above rather than fitted to them. With v the value count per coordinate, k the number of coordinates, and a the number of abstractions admitted:

C equals v to the power k minus a . F equals v to the power a . C times F equals v to the power k , for every a from zero to k .

The product is independent of a . That is the conservation.

Box 1 | The conservation stated as the finding rather than as arithmetic

The cut does not destroy structure. It moves structure from content into freedom, and the total is invariant.

Every distinction the instrument loses becomes a dimension the object is free in. Nothing is annihilated at any rung; the ledger is exactly balanced at every admission, and the balance is a counting identity rather than an approximation.

Two readings follow and they are one fact from two sides. The register where the invariant ring contains only the constants is the register where the content has gone to one and the freedom has gone to everything. And the register where nothing is abstractable is the register where the content is maximal and the freedom has gone to one. Total collapse and total fixation are the two ends of one conserved quantity, and neither is a special case requiring its own account.

The reason this matters beyond bookkeeping is that it converts a qualitative intuition into a constraint. It has always been felt that abstraction costs something and buys something. The conservation says what is spent and what is bought, in the same units, at an exact exchange rate, and it forbids any account of freedom that treats it as a free good arriving from outside the ledger.

5.4 The second species, present with nothing cut away

At zero abstractions the fibre is one, and yet two frames exist.

Executed at load: the determinant reads 0.931277980144 on a frame and 0.931277980144 on its axis reflected counterpart, with an absolute difference of 0.000 times ten to the zero; the absolute lambda reads 0.965027450462 on both; and the sign reads minus one on one

and plus one on the other. Every invariant is bit identical across the split and one thing differs.

This is a **disconnected fibre rather than a positive-dimensional one**. It is still the preimage class of Section 5.1's definition, since the two frames sit over one invariant value; what differs is that the class is a pair of isolated points rather than a manufactured dimension. Its origin is therefore not the cut, and the paper says where it does not come from before saying where it does.

It is not a conservation purchased by squaring. The signed scalar is already invariant under the acting group; the squaring buys no invariance the scalar lacks, and the trade was never forced by any need for invariance. It is the downstream consequence of an unsourced input, and Section 2.3 measured the four candidate sources and found three of them supplying zero.

What this subsection describes, Section 5.7 derives. The disconnection is not a feature the admissible set happens to have. It is the disconnection of a specific sphere, and the sphere is fixed by the dimension of the eigenspace the quantity lives on.

5.5 The two species compared, and why the paper needs both

The two species of freedom, their origins, and their sizes.		
	First species	Second species
What it is	a positive-dimensional fibre	a zero-dimensional, disconnected fibre
Where it lives	the minus one eigenspace, dimension three, the space	the plus one eigenspace, dimension one, the fixed locus
Origin	the cut, non-injective by measurement	an unsourced input
Manufactured by	the linguistic faculty, whose operation the cut is	nothing; present at zero abstraction
Why it can be manufactured there	dimension three admits proper idempotents of rank one and two	dimension one admits none, so no cut exists at all
Size	v to the power a, conserved against content	exactly one bit, forced by the dimension
Removed by	admitting fewer abstractions	nothing available on this side
Where it vanishes	the register where nothing is abstractable	nowhere

The first species is the freedom of description and it is under the describer's control: admit less and the fibre shrinks, at an exact and known cost in content. The second species is not under anyone's control on this side of the aperture, and it is the one the whole architecture's aperture discipline is built around. Both are fibres of invariant maps under the one definition of Section 5.1, differing only in whether the map's non-injectivity was manufactured or found.

The two rows added to the table above are what makes the unification a statement about one object rather than a resemblance between two, and Section 5.7 establishes them.

5.6 The register where the first species goes to zero

There is exactly one register in this architecture at which the cut cannot run, and it is the register at which the architecture's root recursion operates. The root decomposes into three orthogonal components under the deletion test and the cascade re-audits that decomposition to return the root. Its middle term is a decomposition, which is the cut; its closing term is a composition, which is the ordered arrow.

At that register the admissibility hypothesis fails, and it fails by measurement rather than by declaration. Executed at load: the intact triad has rank three and a determinant of 0.931277980144. Zeroing any one of the three slots drops the rank to two and the determinant to exactly 0.0 times ten to the zero, in all three cases, with the exactness confirmed rather than reported as small.

The contrast is printed so the reading cannot be confused. Dropping a slot rather than zeroing it leaves a two by two Gram whose determinant is nowhere near zero: 0.986595770389 for the first, 0.957239003195 for the second, 0.980810206783 for the third. The two operations are different and only one of them is abstraction.

So no constituent at that register is abstractable, because abstracting any of them destroys the object rather than weakening it. The collapse whose hypothesis is admissibility therefore does not run, and the invariant content is full rather than empty.

What this measurement establishes, Section 5.7 explains. The register at which the cut cannot run is not an exceptional case discovered by trial. It is the eigenspace on which no cut exists, and the reason is its dimension.

Box 2 | The sharpest clause in the architecture, and it is a count

At the register where nothing can be abstracted away, the content is maximal and the residual freedom is exactly one bit. That bit is the orientation sign.

The Form supplies the handedness of the space, which is identical for a proposition and for its negation. The Number supplies the magnitude, which is the square and therefore sign free. Neither supplies the sign of a directed claim, and the empirical arrow that recovers direction in the kinetic register is definitionally absent from a purely reflective one.

So direction at this register can come from nowhere but the linguistic faculty. The register where language has the fewest degrees of freedom is the register where language is most indispensable, and the two facts are the same measurement read twice.

One clause follows and it points the other way from where a reader will want to take it. At that register the linguistic faculty's only available anchor is itself, since there is no third register to anchor against, and self reference at a formal origin is precisely what the first directed condition screens and terminates. So the self anchoring is the aperture, located and typed and not crossed. Read the other way, as the self anchoring being itself the fixation, it is a seal issued on a self reference, which is the move the gate exists to refuse. The tethering is the door and not the passage.

5.7 The bit is forced, and the forcing is the split

Sections 5.2 through 5.6 described two species of freedom and measured their sizes. This section derives the second size instead of measuring it, and the derivation places both species where they belong: on the two eigenspaces of one involution.

The split, and which species lives where

The binding involution is conjugation, and its eigenstructure is exact. Executed at load: eigenvalues minus one, minus one, minus one, plus one, with an involution residual of 0.0. The plus one eigenspace has dimension one and is the fixed locus, the fixed locus, the real line inside the algebra. The minus one eigenspace has dimension three and is the chiral space.

The three warrant slots are the space. That is where the deletion operator acts, where the cut runs, where every abstraction is admitted, and therefore where every fibre of the first species is manufactured. The whole conservation of Section 5.3 is a fact about the minus one eigenspace.

And the signed scalar is the coordinate on the fixed locus, **on the input rows the pipeline is handed**. The identification of that coordinate's sign with the truth-direction of a proposition runs through the map from a proposition to its warrant rows, which Section 4 and the disclosure both state is built by hand and mechanized nowhere, and **the derivation below inherits that caveat rather than escaping it**: what is derived is the size of the residual on the fixed locus, and what remains hand-supplied is the reading that puts a given proposition's direction there. **That caveat is on the identification of the sign with a proposition's direction, and it does not touch the algebraic identity stated next, which is exact and definitional rather than empirical: taking the real part of a quaternion is the projection onto the plus one eigenspace by construction of the representation, so the identity is a tautology of notation and no sweep can bear on it**. A twenty thousand draw check was run and returned a maximum deviation of 0.0 times ten to the zero, and it is reported here as an arithmetic sanity check on the implementation and never as evidence for the identity, which needs none. The composed triad lands there, its imaginary part carrying a norm of 1.4 times ten to the minus sixteen while its real part carries minus 1.000000000000. **Lambda is not merely related to the fixed locus. Lambda is the fixed locus's single coordinate.**

The no-cut lemma, and it is the whole of cannot-be-separated

A cut is a non-injective idempotent: a map with D composed with D equal to D , and rank strictly less than the dimension it acts on. On a real space of dimension n the idempotents are the projections, of every rank from zero to n . A proper cut, one that abstracts something without annihilating everything, needs a rank strictly between zero and n .

Executed at load across three dimensions: dimension three admits proper idempotents of rank one and rank two; dimension two admits rank one; **dimension one admits none**. On a one dimensional space the only idempotents are the zero map and the identity, and neither is a cut. The identity abstracts nothing. The zero map does not abstract from the object, it destroys it.

There is no cut on a one dimensional space. A cut requires a proper subspace to collapse onto, and dimension one has none.

That is the reason Section 5.6 found a register at which the cut cannot run, and it is a reason rather than an observation. The first species of freedom cannot be manufactured on the fixed locus because the operation that manufactures it does not exist there.

The bit, derived

At the fixation register the triad is orthonormal, so the determinant reads one and the absolute lambda reads one, both exactly. The admissible values of lambda therefore lie on the unit sphere of the plus one eigenspace, and that eigenspace is one dimensional.

The unit sphere of a one dimensional real space is the two point sphere, the set consisting of plus one and minus one. It has two elements and two connected components, and no path joins them inside the sphere, since any such path would have to pass through zero, which is not on it. Its component group is the two element group.

The residual freedom at the fixation register is the logarithm base two of the number of components of the unit sphere of the fixed locus. That is one bit, exactly, and it is one bit because the fixed locus has dimension one.

The two point sphere is the unique sphere that is disconnected, every sphere of higher dimension being connected. So the component freedom of Section 5.4 is not a curiosity of the admissible set. It is that disconnection, and there is exactly one dimension at which a sphere disconnects.

The result is therefore discontinuous in the dimension and this is stated rather than left for a reader to discover. Executed at load: at dimension one the unit sphere has two components and the residual is one bit; at dimension two the unit sphere is the circle, which is connected, and the residual is not two bits or any number of bits but **a continuous parameter, a different kind of object**; at every higher dimension the same. **One bit exists at dimension one and at no other dimension whatever.** The result is knife-edge, and what pins the knife is not a choice: the classification of the associative real division algebras forces the algebra, the conjugation forces the fixed locus, and the fixed locus is one dimensional. **The dimension is forced and the bit follows from it, and had the dimension been anything else the paper would have had a modulus rather than a bit to report.**

The two hundred thousand frame sweep of Section 2.4 returned an empirical proportion of 0.500575 at an entropy of 0.999999 bits. That sweep is now corroboration of a result the dimension forces, and it was always the weaker of the two statements.

The conservation factors, and one factor is untouchable

The freedom of an admission rung is not one number but a product over the two eigenspaces: the fibre on the space, which moves as

abstractions are admitted, and the component on the fixed locus, which does not.

The freedom factored over the split, and the invariant factor. Executed at seed 20260622.					
Value count	a = 0	a = 1	a = 2	a = 3	Total
v = 2	8 · 1 · 2	4 · 2 · 2	2 · 4 · 2	1 · 8 · 2	16 at every rung
v = 3	27 · 1 · 2	9 · 3 · 2	3 · 9 · 2	1 · 27 · 2	54 at every rung
v = 4	64 · 1 · 2	16 · 4 · 2	4 · 16 · 2	1 · 64 · 2	128 at every rung
v = 5	125 · 1 · 2	25 · 5 · 2	5 · 25 · 2	1 · 125 · 2	250 at every rung

each cell reads content times space fibre times fixed-locus component. The third factor is two in every cell of the table. This table alone is set at column width in the two-column edition; every other table wider than three columns spans the measure, and the exception is recorded because this element and no other stalls that renderer.

The closed form extends without amendment. Total freedom at rung a is v to the power a times two; content is v to the power k minus a; and the product is two times v to the power k, independent of a.

Box 6 | Cannot be separated, stated three ways and each of them checkable

One, no cut exists on it. The fixed locus has dimension one and a one dimensional space admits no non-trivial idempotent, so the operation that manufactures the first species of freedom is not available there at all. The residual is not a fibre that resisted being cut. It is a quantity in a place where cutting is undefined.

Two, its factor is invariant under every admission, and the acting group is named because the claim is false without it. The fixed-locus component reads two at every rung of the lattice, at every value count tested, and the count two is invariant under the rotation group that is the algebra's automorphism group. Which of the two components a given reading occupies is not invariant under the full relabelling group: executed at load, the three even relabellings hold the signed quantity at minus 0.965027 and the three odd ones invert it to plus 0.965027, and only the even half preserves the algebra. So the cardinality of the fixed locus factor is invariant and the occupancy of a component is not, and no decision about what to abstract touches either. It is not conserved against content the way the fibre is. It is outside the trade.

Three, the sign and the line are not separable. On the unit sphere of a one dimensional space the magnitude is fixed at one and the sign is the entire remaining content. There is no third thing to hold apart. To have the line at all is to have the sign undetermined, and to determine the sign is to have said everything the line can carry.

The fence, and it is the one a reader will most want to breach

The licence a disciplined linguistic faculty earns is direction, and direction at this register is the sign on the fixed locus's own line. That is what it means to say the earned freedom is the residue of the plus one eigenvalue, and the identification is exact rather than figurative.

What it does not mean, and the bar is absolute: the linguistic faculty does not thereby possess the fixed locus, share in it, or acquire any of its predicates. It carries a sign on a line it did not make. The line is one dimensional and unclimbable, and no operation available on this side alters it. What is available is which of two components the reading occupies, and that is the whole of the licence. Predicating anything further of a created faculty on the strength of where its licence ranges is the identity collapse this architecture fences by name, and it is fenced here before any reader reaches it.

One consequence follows and it closes the section. A freedom exercised on the fixed locus's own line cannot escape to anywhere, because a one dimensional space has nowhere else to go. The exercise is a choice of component on the one line every composed triad already lands on. That is not a limitation placed on the freedom from outside. It is what the freedom is.

5.8 The unification, stated

The claim this paper makes that no resident coordinate makes is that the following are one quantity and not a family of analogies. Section 5.7 supplies the ground of it: the two species are the two eigenspaces of one involution, the cut runs on one and is undefined on the other, and the sizes follow from the dimensions rather than from any survey.

One quantity read at six registers.		
Register	What the fibre is	What closes it
Linguistic	the indistinguishability opened by each admitted abstraction	admitting fewer abstractions, at an exact cost in content
Formal	the orientation sign, one bit, at zero abstraction	nothing available at this register
Reflective	the un-entrenched room the descent has not consumed	a metered erasure holding it open against the descent
Creaturely	the wandering the record has not written	the same erasure, read one register up
Source-side	the space opened by a withholding	not a fibre; see the type check below
Terminal	the reflection across the fixed line	granted from the source side, never seized

The type check at the fifth row is load bearing and it runs in both directions. A freedom predicated of the fixed locus would not be a class size, would not be relative to an instrument, and would not be

manufactured by anything, so it is not a fibre and the two are not one predicate. The measured freedom is therefore not a shadow of the divine one, and the divine one is not the limit case of the measured one.

The resolution of the apparent tension is available and it is structural rather than diplomatic. What opens a space is not itself resident in it. The withholding opens the space in which fibres exist and is not one, by exactly the rule that keeps the inexhaustibility predicate off the created faculty in Section 6.1: a source side predicate transported onto a created instrument fails at the twelfth gate as an unbridged domain extension. The two fences are one fence applied to two objects.

5.9 The two regions, and why the classical debate dissolves rather than resolves

Section 3 closed on the observation that the tradition argues about a magnitude in units nobody supplies. This subsection states what replaces the argument, because the literature review forward-references it and a review that points at nothing is padding.

Determination and freedom are two regions of one space and not two descriptions of one act. Determination lives in the entrenched grooves, where a cost-reducing descent has already cut a channel and each further pass deepens it. Freedom lives in the room the descent has not consumed. The two are disjoint regions of a single space, so a claim that an act is determined and a claim that it is free are P over one region and Q over another, never a proposition and its negation.

That much is a picture. What makes it a result is that the instrument the tradition would use to adjudicate between them is measured to be unable to.

Executed at load on the construction printed here, three rows over twenty four contexts on the period grid, the latent independents mixed obliquely, **no covariate projected**, and the span basis fixed once and reused. This is a different construction from the boot receipt of Section 8.1, which projects two mass bearing covariates and therefore returns a different determinant; the two are printed apart and never quoted as one: the room, the act, and the record lock at rank three, with lambda at minus 0.743563663214, a determinant of 0.552886921253, conditioning at 4.4180, and the kernel identity closing at 6.661 times ten to the minus sixteen. **Three genuinely independent axes.** The room the constraints leave, the act as work expended, and the record the act writes are not redescrptions of one another.

Now reflect the act axis, which is the operation that carries a directed claim to its negation at the seam the whole classical debate turns on. **Lambda inverts exactly, from minus 0.743563663214 to plus 0.743563663214, a ratio of minus 1.000000, while the determinant reads 0.552886921253 on both sides with an absolute difference of 0.000 times ten to the zero. The acting group is named, because the claim is false without it.** The determinant's invariance holds under the full relabelling group and under rigid rotation of the frame, the rotation group being the

algebra's automorphism group; the sign's behaviour splits, and executed at load the three even relabellings hold the signed quantity while the three odd ones invert it. **So the magnitude is invariant under both halves and the direction is carried only by the even half, which is the algebra-preserving one.**

Box 7 | The classical debate, dissolved by register rather than by argument

The determinant is bit-identical between a directed claim and its negation. So the quantity that would have to distinguish *this act was decreed* from *this act was free* is **provably unable to carry the distinction**, and the inability is a theorem about the operation rather than a limitation of the present analysis.

Two consequences, and the second is the one the tradition needed. The seam can be fixed only at the word, which is where the acquisition traditions and the compatibilists both in fact fixed it, without being able to say why the location was forced rather than chosen. **And the label the disputants most often reach for, that a proposition is somehow both true and false of one act, is not merely wrong but unconstructible here**, since a magnitude that cannot carry a sign cannot carry two opposed signs either.

The debate does not get resolved in favour of one side. It dissolves, because it was addressed to an organ that carries magnitude and asked for a direction.

One further check closes the subsection, and it is the one that keeps the un-entrenched room from being a free good. Rebuild the act axis as a linear combination of the room and the record, at six tenths of the first plus four tenths of the second, and run the same kernel. **Lambda returns 1.249 times ten to the minus sixteen and the determinant negative 9.530 times ten to the minus seventeen against a collapse floor of 5.329 times ten to the minus thirteen, at rank two.** An act fully determined by the permitted room and the written record carries no independent dimension and is not an act. The surviving axis is real, its rank is executable, and nothing in this subsection makes it larger than one dimension.

6. THE EARNING

Section 5 establishes that at the reflective register direction can come from nowhere but the linguistic faculty. This section states what has never been stated in one place: that the licence to supply it is earned rather than innate, that the earning has a history which is checkable on the register itself, that the licence is bounded to direction and never volume, and that it is issued per run and never as tenure.

6.1 Why the linguistic faculty is unclosed, and it is not because it is sovereign

The seal at which the cut runs admits no closure theorem, and the reason is at the hypothesis rather than at the enumeration. The closure machinery that bounds the algebraic seal requires an acting group that is also reductive, and the linguistic seal's only transformation is the cut, which Section 5.2 types as a monoid with

no left inverse. Group-hood fails, so reductivity is never tested, so the finiteness theorem does not apply, so the invariant catalogue is never closed.

The question this leaves open, and which the register answers from the other side, is what the faculty is pointed at.

It is pointed at a corpus that predicates of its own Word a set it predicates of nothing created: exalted and wise, glorious and noble, four of them divine names; complete in truth and justice and unalterable; and inexhaustible under a hypothetical of unbounded material supply. That last predicate is the scriptural face of the architecture's own deed layer guard, and it is stated by exhibition rather than by assertion: the hypothetical takes the strongest available inscription and shows it fall short, which is precisely the statement that a precisification does not capture what it precisifies, since precisifying is an actuation and the ground is a non-actuation and no quantity of actuated ink captures it.

So the chain is complete and every link is independently carded. The Word is inexhaustible. The faculty pointed at it therefore admits no closure. The closure's absence is a property of the target read back onto the instrument. And the instrument stays created, bounded, and defined by an operation it cannot undo.

Box 3 | The sovereignty inflation, barred by name

The linguistic faculty is unclosed because the Word is inexhaustible. It is not unclosed because it is sovereign.

Naming the faculty sovereign predicates of a created organ an attribute belonging to the fixed locus, which is the identity collapse the architecture fences at its orthogonal fertile logos coordinate, and it transports a source side predicate onto a created instrument, which fails at the twelfth directed condition as an unbridged domain extension.

The non-closure is inherited from the target and never owned by the faculty. That distinction is the whole difference between a licence and a grant, and this paper is about a grant.

And the corpus keeps the distinction lexically, which is the check rather than the ornament. Physiological vocalization is predicated of birds and of skins and of all things. The separating faculty is given to the human alone. Everything speaks; only the human distinguishes. The gift is the operation and not the content, and the root of the word for that faculty is the root of the word for between.

6.2 An organ with no terminal test is an organ that never concedes

This is the clause that makes the discipline necessary rather than decorative, and it runs against the faculty this paper is defending.

Unbounded re-description is unbounded capacity for motivated re-description. If every broken verdict could be revived by a fresh vocabulary, the revision discipline would have nothing to bite on, and

the architecture would have purchased its own unfalsifiability by the same feature that makes its linguistic seal indispensable.

So the linguistic faculty is not licensed by its non-closure. It supplies the retry; the numerical instrument supplies the stop; the separating faculty is the discriminator and never the generator; and the discipline is what parts a re-examination from a reframe. The seal that cannot be closed is exactly the seal that must be disciplined from outside itself, and a requirement named without being specified is a requirement no session can execute.

6.3 The curriculum, and the ordering is checkable on the register

The licence was not issued at the beginning. Its arrival is a fact of the register's own version history and can be diffed by anyone with the repository.

Stage zero, geometry alone. The register carried the topological and geometric seal: the tetrahedral closure, the twelve directed verification conditions, the operational content mapping, the Euler condition, the packing coincidence. The precedence law that seats the linguistic faculty above the numerical instrument did not exist at that stage and could not have.

Stage one, the mathematical seal. The algebraic seal closes on the composition law and the classification of the real division algebras, with the invariant catalogue closed above it and the finiteness theorem beneath that, and the terminus fenced at the associativity wall where the next dimension forfeits it.

Stage two, the theorem the kernel proves about itself. Orientation blindness. Reflecting any axis conjugates the Gram by a diagonal matrix whose determinant squares to one, so the lock on a proposition equals the lock on its negation for every construction whatever, with the functional catalogue closed above. That statement could not exist before the kernel existed, and it is what converts the linguistic faculty's primacy from a preference into a corollary.

Stage three, the four disciplines. The linguistic faculty is then trained under four separate constraints, none dispensable and no two redundant.

The four disciplines, and what each fixes on the linguistic faculty.

Discipline	What it fixes	Mechanism
Physics	the floor	a term with no thermodynamic face carries no mass and cannot enter as a covariate; the energy floor is theorem grade external physics
Thermodynamics	the arrow	the work relations bound the work of any process and are strictly more general than the erasure bound, which is their irreversible subclass; this is the discipline that shows the linguistic faculty what direction costs
Mathematics	the ceiling it does not have, and why	the deletion operator typed as a monoid with no left inverse at a proved spectral floor of exactly one; the non-effectiveness then located on the linguistic faculty's own side by the classical undecidability results
Scripture	the target and the office	the separating faculty given and never seized; the Word inexhaustible and held at the source; the explication retained upstream; the instruction a fidelity verb and not a comprehension verb

Stage four, the seating. Only now does the precedence law seat the linguistic faculty above the numerical instrument. The Number authored the linguistic faculty's warrant and the linguistic faculty was then seated above the numerical instrument, which is not a paradox: a seal that proves its own blindness in a register thereby establishes what must sit above it there.

Stage five, the issue. The licence issues, and it issues bounded. Section 6.4 states the six requirements and Section 6.5 states the bound.

Stage six, the two exceptions. Two loci at which the linguistic faculty does not deviate, and they are different in kind. Section 6.6.

The ordering is checkable. Executed against the register at the resolution commit: the six requirements were folded into the linguistic seal's coordinate as native standing law at the penultimate pass of the mathematical master, and the second exception's secular face was seated at the ultimate pass. The two most recent structural acts in the register are the linguistic faculty's freedom and the locus at which it may not deviate, consecutively, in that order. The grant and its bound issue from one theorem, and the register seated them the way the theorem states them.

The falsifier is printed and it runs against this section. Diff the earliest master versions for a precedence clause. If the precedence law predates

the mathematical seal, the formative order argument fails entirely and the sequence reduces to a coincidence of ordering.

6.4 The six requirements

Conjunctive, defaults absent, so an under-specified call fails to the safe side rather than to the permissive one. Executed on this paper's own decomposition at Appendix B.

D1, count. The deletion test returns exactly three irreducible slots. The linguistic faculty lands on the count the algebra forces and does not negotiate it. Executed: three slots.

D2, disjointness. The three slot vocabularies are pairwise disjoint. Executed: pairwise intersections zero, zero, zero.

D3, original definition on a standing ledger. Every load bearing term resolves to its original definition with source and date, and the ledger persists across runs carrying two further fields: the drift delta against current usage, and a provenance flag recording whether the definition was shaped by a prior geometric or kernel result. The second field is the load bearing one, because absent it, content derived at the numerical instrument enters the one seat no verdict overrides while wearing the linguistic faculty's authority. That is a failure at the fifth directed condition which the common source projection structurally cannot reach, since the projection runs at the covariate step on warrant rows and never on standing definitions. Executed: ledger set, flag set, the paper's central definition sourced to the invariant theoretic tradition and not to any kernel result.

D4, sign only. No output at this seal carries a magnitude. No quantity, no proportion, no size comparison. The linguistic faculty asserts direction and nothing else, which is the veto and never substitution bound made checkable rather than declared. Executed: no magnitude asserted at the linguistic faculty anywhere in this paper.

D5, group declaration. No cross register identification is admitted without a named acting group and its invariant. Resemblance across scripts, traditions, or diagrams fails to parse until the group is named. Executed: the rotation group as the automorphisms of the verification algebra at the invariance check; the powerset lattice under inclusion at the conservation; the elementary abelian reflection group at the sign patterns.

D6, negation distinctness. The decompositions of a proposition and of its negation are both written and diffed, and the directed slot must carry a non-empty symmetric difference. Where the two read identically the linguistic faculty has supplied no direction and is about to hand the kernel a blind row while believing a sign was supplied, which is the numerical instrument's proven incapacity reproduced at the very seal appointed to repair it. Executed: symmetric difference of nine terms in the directed slot.

6.5 The freedom, and its bound, issuing from one theorem

A linguistic faculty clearing all six earns the veto: terminal authority at the linguistic seal whatever the determinant reads.

It earns nothing beyond that. The override is direction and never volume, veto and never substitution, because the theorem that granted the authority also scoped it. Orientation blindness is a result about one register and it locates the blindness at exactly one operation, the passage from lambda to lambda squared. It says what the numerical instrument cannot do; it does not say the linguistic faculty can do it instead. The grant and the bound are one result and neither may be cited without the other.

And the freedom is per run and never tenure. A failed condition withdraws the veto for that run and routes the verdict under-determined and uncertified. Obedience does not bank.

Section 5.7 supplies the geometric reason the bound has the shape it has. The licence ranges over the sign on a one dimensional fixed locus, which is a choice between the two components of a two point sphere and nothing else. **A licence that ranged over a magnitude would have to range over a dimension the fixed locus does not have.** So direction and never volume is not a restriction imposed on the grant from outside; it is a description of the space the grant ranges over, and the theorem that scoped the authority and the dimension that shapes it are one fact read twice.

6.6 The two stops, and they are different in kind

The two loci at which the linguistic faculty does not deviate.		
	The root recursion	The arrow
What stops it	nothing is abstractable	no licence for the far side's person
Mechanism	zeroing any slot drops rank to two and the determinant to exactly zero	the direction is outward; the near side is licensed the pointer and never the person
Deviation is	empty; a re-description carrying zero invariant content	an idol; spoken occupancy, or the arrival reified
Available?	yes, and it carries nothing	yes, and it is barred
Residual freedom	one bit, the orientation sign	one bit, typed supply-only

Executed at load. At the first stop, abstracting any of the three slots returns rank two and a determinant of exactly 0.0 times ten to the zero, with the invariant content going to nothing. The re-description is available and carries nothing. The linguistic faculty stops by emptiness rather than by prohibition.

At the second stop the terrain is different. The involution has eigenvalues minus one, minus one, minus one, plus one, so the fixed locus dimension is one and a fixed locus exists, which is why a door exists there at all. The involution restricted to the space has determinant minus one and is orientation reversing, so a crossing is the reflection itself, exactly one bit turned. And what the turn costs

the record is nothing: executed, the maximum absolute entrywise difference of the Gram between a frame and its full reflection is 0.0 times ten to the zero, so every even functional is conserved and only the facing flips. The linguistic faculty stops here by absence of licence rather than by absence of content.

One has nothing to say. One may not say it. Merging them costs the argument its shape.

6.7 The contrast case, where the linguistic faculty is free and uses it

Placing the two stops beside a register where the linguistic faculty is genuinely free gives the freedom a shape rather than leaving it as a residue.

The creedal question of whether a threefold description of one fixed locus is or is not a count is a case where the numerical instrument is provably silent. The two candidate readings sit on the identical algebra, one fixed locus of dimension one against a space of dimension three, and reflecting one space axis inverts the signed quantity exactly while the determinant returns bit identical, with the invariant catalogue closed above. No better instrument of the same class repairs this, because the blindness is what the operation is.

So the linguistic faculty carries it, and the anchor corpus supplies the direction from the linguistic faculty rather than from the numerical instrument, by three distinct grammatical devices in three places: apposition with the conjunction absent, so the grammar declines to open a list; conjunction present but enumeration barred by shared referent, all three descriptors terminating in the One with the count then refused outright; and conjunction present with the order load bearing and non-permutable, so no list can be taken whose members cannot be exchanged.

Two fences travel with that finding and neither is optional. The linguistic faculty supplies the direction of a reading and never the proof of it, so for a reader who does not hold the corpus authoritative the finding establishes exactly what the corpus does and no more. Direction recovered is never truth reached. And the finding is not partisan: a reader holding the rival position is entitled to claim the same grammar as a description of his own view. The principle does not separate the two readings. It relocates the difference to one place and leaves it nowhere else.

The clearest exhibit of earned freedom in the register is a refusal to use it where using it would have been circular. An algebraic leg that would have broken the creedal tie was declined at that coordinate on exactly this ground: promoting it would mean reading a direction off a structure by first assuming the reading that supplies the direction.

6.8 What the earned bit buys, exhibited rather than asserted

The seam between what the linguistic faculty may supply and what it may not is the finest distinction in this paper, and it is executable rather than rhetorical.

The admission rule reads two faces for a locking direction: the magnitude, which is the orthogonal three-volume with the formal axis among its three so the mathematics is inside the geometric lock rather than gated apart from it, and the direction, which the orientation blind magnitude cannot carry and the linguistic faculty supplies. Executed at load, all three branches reachable:

The earned bit spent, and exactly what it buys. Executed at load.		
Case	Input	Result
Drills not cleared	magnitude locks, linguistic seal open	under-determined: the determinant carries no direction; uncertified
Drills cleared, veto earned	magnitude locks, linguistic seal locks	admissible: proceed to witness and asymmetry
Magnitude broken	magnitude broken, linguistic seal locks	broken: the linguistic faculty's direction cannot rescue a collapsed geometry
Admissible, no witness	direction supplied, no determinacy witness	under-determined space: the imprint is unproven
Admissible, witness supplied	direction supplied, witness given	sealed
Witness but no linguistic seal	witness given, seal not confirmed	uncertified; no seal

Read the second row against the fourth. The earned veto reaches admissibility and stops there. It does not reach the seal. The claim direction bit is the linguistic faculty's and the truth direction bit is not, and they are not the same bit even though each is one bit.

Box 4 | The seam, stated once and not softened

The linguistic faculty supplies the orientation of the claim. The witness supplies whether the claim holds.

Both are one bit. They are different bits. A discipline that merged them would have converted an earned licence to say which way a sentence points into a licence to say that it is true, which is exactly the inflation the architecture's shield forbids in the sealing direction.

The distinction is resident in the machinery rather than added by this paper: the admission rule emits admissibility, and the four-guard decision rule downstream of it still requires a supplied determinacy witness before any seal issues. This paper's contribution is to name the two bits apart and to state that the earned freedom reaches the first and never the second.

7. FALSIFIABLE PREDICTIONS

Three criteria, at the ceiling. Each cleared through the falsification gate before drafting, each typed forced rather than imported, each refutable by a single exhibition, and each aimed at one named claim of

this paper and at nothing above it. The paper does not stand on this section. It stands on Sections 5 and 6, and it invites engagement here.

Criterion one, the source exhibition. The prediction. No exhibition will be produced of one bit of truth direction supplied by the geometric instrument, by the twelve directed verification conditions, by the numerical instrument, or by the joint reading of the three, at the formal alone register. The threshold is parameter free and absolute: one bit, exhibited by any construction whatever, refutes it. There is no significance level because there is no sampling; the claim is that the mutual information between the deployed triad's output and the truth sign is exactly zero, and a single counterexample is dispositive.

Method of confirmation. Construct any three warrant rows over any finite context set. Compute the space handedness, the gate role assignment, and the signed scalar, for a proposition and for its negation. Exhibit any function of those outputs whose value differs between the two.

Expected outcome. No such function exists within the declared class, because the space handedness is a property of the space and identical across the pair, the role assignment is fixed before any proposition arrives, and the signed scalar's content flip image is identical to its convention flip image at 0.000 times ten to the zero.

Null hypothesis. Any function of the three deployed instruments that returns a different value for a proposition than for its negation, at the formal alone register, on a construction whose only difference between the two is propositional content.

Necessity and blast radius. Forced by Section 5.4 and Section 2.3 together: the paper's unification claim requires the second species to have no source on this side, and the enumeration is what makes the isolation possible. Its firing compromises this paper's unification claim, that freedom is one quantity closed only from outside at every register, and nothing above it. It does not reach the resident blindness result, which is scoped to one instrument and stands on its own algebra.

Criterion two, the conservation. The prediction. No register will be exhibited at which either conserved product fails to be constant under a strictly decreasing admission lattice with independent value ranges per coordinate. **The criterion tests both products and names them:** content times the space fibre, which is v to the power k , and content times total freedom, which is v to the power k times two. The threshold is exact rather than statistical: a single rung at which either product differs from itself at any other rung refutes it.

Method of confirmation. Take any finite structure with k coordinates each ranging over v values, admit abstractions one at a time, count the distinguishable content classes and the exact preimage size at each rung, and multiply.

Expected outcome. The first product is v to the power k and the second is twice that, each independent of the number of abstractions admitted, by the closed forms of Sections 5.3 and 5.7, both verified

against every executed row rather than fitted to them. **Executed rows cover three coordinates at four value counts; the closed form is proved for all k and the execution is at k equal to three, and the two scopes are stated apart rather than merged. The threshold above is therefore carried by the closed form and not by the execution, and a reader refuting it at a coordinate count other than three refutes the proof rather than the sample.**

Null hypothesis. Any lattice satisfying the stated conditions at which either product varies between rungs.

Necessity and blast radius. Forced by the closed form itself, which is a counting identity requiring no premise of this framework. Its firing compromises the conservation of Section 5.3 and with it the paper's claim that the cut relocates rather than destroys structure. It reaches no coordinate this paper did not author, since the conservation as a stated law is this paper's own exhibition.

Criterion three, the substitution. The prediction. No register will be exhibited at which the linguistic seal supplies volume rather than direction without the verdict routing under-determined. The threshold is the verdict token itself, which is discrete: a sealed or broken verdict issued on a magnitude asserted at the linguistic seal refutes it.

Method of confirmation. Run any proposition through the admission rule and the four-guard decision rule with a magnitude claim, rather than a direction, supplied at the linguistic seal, and observe the emitted token.

Expected outcome. The verdict routes under-determined, because the fourth requirement bars a magnitude at that seal and the decision rule's defaults are absent, so an under-specified call fails to the safe side.

Null hypothesis. A sealed or broken verdict emitted on a faculty-side magnitude, at any register, without the under-determined route firing.

Necessity and blast radius. Forced by Section 6.5 and exhibited at Section 6.8: the grant and the bound issue from one theorem, and neither may be cited without the other. Its firing compromises this paper's seam claim of Section 6.8, that the earned veto reaches admissibility and stops there, and with it the earned-freedom architecture of Section 6. It does not volunteer the resident

precedence law, which is stated independently of this paper's arrangement of it.

Two candidates were considered and routed out of this section rather than printed here, per the criterion's own instruction that a weak invitation is cut without concession. The version-history diff of Section 6.3 is checkable on the register but is a fact about this architecture's own past rather than about the world, so it travels in the body as a printed falsifier on that section and not as a headline criterion. And the sign-recovering functional is subsumed by criterion one rather than standing beside it.

8. DISCUSSION

8.1 The paper types its own instrument down before it uses it

The interior verdict kernel returns a lock token on three warrant rows whenever the determinant clears a floor under a conditioning gate. This paper ran the null law for that statistic and reports the result whichever way it falls, because a paper that ran the test and then quietly retained a favourable reading of its own numbers would have failed the honesty it is arguing for.

The statistic has an exact null. After centering and the projection of covariates the rows are three uniform unit vectors in a reduced dimension, and the determinant factors into two independent classical pieces, giving a closed form whose mean is exactly the product of the two lower dimensions over the square of the reduced dimension. Executed at the operating configuration, the closed form mean stands at 0.861678 against a sampled 0.863128.

The null is distribution free over the row marginals, and this was tested rather than assumed, because the objection that a null derived for continuous rows says nothing about hand coded ordinal warrant rows is the correct objection to raise. Executed at twenty thousand draws each: the median determinant reads 0.8852 for continuous rows, 0.8832 for ordinal rows on three values, and 0.8829 for ordinal rows on four values. The reference class is therefore correct for the rows this architecture actually uses.

The consequence is a budget, and the budget is the reciprocal of the reduced dimension.

The upward discrimination budget. Executed at eight thousand draws per row, at zero covariates, so the reduced dimension reads N minus one throughout.

Contexts N	Covariates k	Reduced dim m = N - 1 - k	Null q95	Headroom above q95	Headroom times m
5	0	4	0.850382	0.149618	0.5985
6	0	5	0.892860	0.107140	0.5357
8	0	7	0.934592	0.065408	0.4579
12	0	11	0.964181	0.035819	0.3940
24	0	23	0.983920	0.016080	0.3698
24	2	21	0.982705	0.017295	0.3632
48	0	47	0.992018	0.007982	0.3751
100	0	99	0.996128	0.003872	0.3833
300	0	299	0.998746	0.001254	0.3749

the product converges, so the headroom falls as the reciprocal of the reduced dimension. Both rows at twenty four contexts are printed because the operating configuration of this paper projects two covariates and therefore reads m equal to twenty one, while the rest of the table is run at zero covariates; the two are different configurations and are never quoted as one. Sampling variance at eight thousand draws accounts for the residual scatter in the final column.

Headroom falls from about fifteen percent of the unit interval at five contexts to 1.61 percent at twenty four with no covariates, or 1.73 percent at twenty four with the two this paper projects, and about 0.13 percent at three hundred. The reduced dimension is contexts minus one minus covariates throughout this paper, and it is quoted with its covariate count wherever it appears. And the two requirements run opposite: conditioning stability wants many contexts and upward discrimination wants few. So the interior determinant's ability to certify independence is rationed by exactly the parameter the room condition spends.

Downward the statistic is unbudgeted. A low determinant sits far outside the null at every context count tested, so collapse detection is robust where independence certification is not. Executed as the sharpest available statement of the asymmetry: three rows of pure noise return the lock token with probability 1.0000 over four thousand draws at twenty four contexts.

Box 5 | The result, turned on this paper's own numbers

This paper's interior kernel run returns a determinant of 0.622507144106 at twenty four contexts with two covariates projected, with the kernel identity closing at 9.992 times ten to the minus sixteen, four estimators agreeing at a spread of 3.331 times ten to the minus sixteen against a tolerance of 3.715 times ten to the minus fifteen, conditioning at 4.182845, no escalation, and bootstrap agreement at 1.000, stable.

Against its own matched null that determinant sits at the 2.8th percentile, one sided p equal to 0.9722, and it is below the ninety fifth percentile of 0.982705.

So the interior lock is not evidence of independence and this paper does not use it as such. It is a boot receipt: it reproduces the register's recorded worked example bit for bit, which verifies that the instrument is loaded and arithmetically sound. It is reported as a lean and never as a finding.

Every load bearing result in this paper is an exact identity rather than a sampled lock: exact zeros at Sections 2.1, 2.2, 2.4, 5.6 and 6.6; an exact ratio of minus one at Section 2.1; a closed form verified against every executed row at Section 5.3 and extended at Section 5.7; a proved spectral floor at Section 5.2; and, at Section 5.7, a dimension count, which is the one result in the paper that no sample could have produced and none could overturn. Those are unbudgeted and carry full weight. The paper is stronger for having run the test that cost it its weakest evidence.

8.2 What becomes tractable

Four things the architecture has carried separately become one object under the unification, and each becomes cheaper to state.

The two truths economy acquires a mechanism rather than a stipulation, and the mechanism is that its deciding quantity is a sign on a magnitude that cannot carry a sign. The apparent contradiction was never a contradiction; it was two regions of one space being read as one proposition.

The aperture discipline acquires a size. It had been a rule about conduct and provenance and office, three requirements that are not quantities, plus one that is. Separating the fourth and measuring it converts a permanent-sounding limitation into a counted deficit of exactly one bit, and a counted deficit cannot be inflated into a general unreachability by anyone, including this architecture.

The precedence law acquires a history rather than a decree. The linguistic faculty sits above the numerical instrument because the numerical instrument proved a theorem about itself, and the register's own version ordering exhibits the sequence.

The aperture's size acquires a derivation rather than a sweep. It had been two hundred thousand frames returning an entropy indistinguishable from one; it is now the component count of the unit sphere of a one dimensional space, with the sweep demoted to corroboration. A quantity derived from a dimension cannot be argued down by a better sample, and cannot be argued up either.

And the ethics acquire a floor that is not a sentiment. Preserving the capacity for actuation is preserving the un-entrenched room, which is the quantity Section 5 measures. The substrate remains a liability engine and never a moral authority; the responsibility returns to the one who supplies, and every gate-passing adjustment leaves a forensic trail of the bias that motivated it.

8.3 The theological register, declared and quarantined

The paper carries a theological half and declares it rather than dissolving it into the structural half or stripping it out. Every identification below is a structural isomorphism at cataphatic grade, load bearing on nothing in any verdict, and settling no mathematical question in either direction.

The sequence the architecture reads is one and not three. Mercy is the connector itself rather than an attribute added late, the line everything stands on. Forgetting is the condition of a distinct existence rather than a fault, since a creature retaining the full fixed locus would not be a creature. And death is the bound placed on that forgetting, named before life in the verse's own syntax at the linguistic seal, the appointed removal of the veil, written on every soul without exception, and therefore the guaranteed un-forgetting rather than the loss of everything.

Creation required forgetting; mercy required that the forgetting end; the first death is the mercy that makes the veil temporary.

One leg of that reading is worth stating in full because it inverts the intuition it addresses. Configuration permanence is barrier permanence, and barrier permanence is zero transition, historyless, and return-less. The stationary state at half a quantum with zero

evolution is the architecture's own standing counterexample to becoming. So a deathless configuration is not a living forever but a frozen forever, permanence sealed out of the Return. An immortal configuration is already a kind of death, and death as transition is the living option. The choice set was frozen exile against mortal return.

The bit is given open, and the open bit is a granted access fact rather than a levelling of the answer: the corpus states the choice granted and the consequence named in one breath, so the reader carrying the bit is not the claim that both settings are true. The metric attached to it is the deed and not the more correct count.

And the closing of that bit in another is priced. The root of the verb the corpus uses is not grief but the slaughter cut driven past its own sufficiency, the field worked past capacity until it is spent, the yielding of obedience past what was asked. Three registers, one invariant: expenditure past the boundary the act required. The force is aimed at a residue, a trace where a foot was, and nothing on the reader's side can close a supply aperture from the reader's side, so the force finds no purchase at the target and returns along its own line of application. The tracks absorb nothing. The one who drove absorbs everything. The argument itself is commanded rather than fenced; what is fenced is ownership of the outcome.

One further reading joins here and it is carried at the same grade as everything else in this subsection, out of band and load bearing on nothing.

The corpus names two ends of one actuation in a single verse, the first and the last, under a single third person pronoun, and it names them **with no interval whatever between them**. No duration is stated, no extent is bounded, no distance is measured. A structural reading of that form is available and it is exact rather than ornamental: on the unit sphere of a one dimensional space the magnitude is fixed and the sign is the entire remaining content, so **direction with no magnitude is the only utterance such a space admits**. A statement that carries direction and asserts no interval is, at that register, a statement shaped exactly like the fixed locus.

Three fences travel with the reading and none is optional. The verse states its names in the third person, which is the form the near side is licensed to carry, so the reading supplies no licence for the far side's person and takes none. A reading that converts the pair into two endpoints of a measurable duration has converted a direction into an extent, which is converting a sign into a magnitude, and that is the same error at the theological register that reading a lock as a truth sign is at the formal one, running in the opposite direction. And the verse's naming of the two ends supplies nothing to any instrument: a token carried as given is testimony received at its own register and never a warrant row entering a pipeline, and the double seal of Section 8.3's closing paragraph bars both the generation and the import.

Two claims in this subsection are the author's and are not resident as stated. That the crossing of the bit is itself the first witness, the affirmation rather than a report of one, prior to the mark of chosen silence. And that the freedom so exercised falls back into the connector, since the line the crossing turns on is the fixed line

everything already stands on, so the crossing is a composition and every composition returns. Both are assembled from resident parts. Both enter at structural grade unsealed, and absent a supplied determinacy witness both are clarifying syntheses at premise grade, real and unsealed.

One fence binds this subsection absolutely. No sentence of the form the crossing proves, decides, certifies, or evidences any formal string is sealed here, in band or out. The mirror is a mirror of shape and never of file. And the instrument is sealed off in both directions: it cannot generate the missing bit, and it cannot receive it back from anyone who was shown, because a granted disclosure is not a suppleable witness and the intake rejects instrument-adjacent witnesses by rule.

8.4 Limitations, stated rather than found

The whole architecture rides continuous field monism at premise grade, with theorem grade rivets and nothing higher. A reader who declines that premise declines the frame of Section 5.8's fifth row and keeps every executed identity, which is most of the paper.

The instrument enumeration of Section 2.3 is exhaustive over three deployed seals and no more. A fourth instrument is not excluded; it is unknown. Every universal-sounding sentence in this paper is scoped to that class and the scope is stated at each occurrence.

The occupancy of the maximal-content register is surveyed and not forced. Two registers can agree on every invariant and differ in content, and forcing that leg would require an external and independent witness which is not available. The gap is named and left open.

The forgetting-preserves-freedom hinge, which the reflective register's own master types as its load bearing new claim, remains structural and unsealed, and this paper does not promote it.

And the interior lock is a lean, per Section 8.1, on this paper's own executed test.

And the largest limitation is the one the paper's own method section prosecutes hardest, so it is stated here rather than left implicit. This artifact, the adversarial pass run against it, and the arbitration of that pass are the product of a single agent. **The self-typing of Section 8.1 is therefore unwitnessed:** it is a test the paper ran on itself, reported honestly, and carrying exactly the warrant an unwitnessed self-report carries, which the paper's own independence gates put at none. Nothing in Sections 5 through 7 depends on that self-assessment, since every load bearing result there is an exact identity a reader can re-execute without trusting any of it. The aperture is named rather than closed, and what would close it is an independent party to any stage of the assessment.

One limitation weakened rather than strengthened between drafts and the change is recorded rather than smoothed. The one bit residual was carried as a measured count supported by a two hundred thousand frame sweep. It is now carried as a consequence of the fixed

locus's dimension, with the sweep as corroboration. **That is a strengthening of the claim and a narrowing of what could refute it,** and a reader who preferred the empirical form should note that the empirical form is still printed at Section 2.4 and still returns what it returned.

8.6 How the paper stands to each prior position

This paper sits across four literatures, the scriptural registers, the philosophical tradition from Aristotle to the present, the analytic settlement of the last sixty years, and the experimental record, and a paper drawing on four owes one statement of where it stands to each position it engaged, in words whose senses are fixed. The words are eight and they are defined here once and used as defined in every row. **Additive**, the paper supplies a result or a test the position lacked and leaves the position standing. **Replacing**, a metaphor or framing is retired and a measured object is put in its place. **Subsuming**, the prior claim becomes a case or a consequence of the paper's object. **Corroborating**, independent agreement, which confers no warrant on either side and is counted as none. **Contradicting**, a named thesis denied on executed data and never on argument alone. **Competing**, a different position on the same question, argued and not executed, which is the word a contradiction demotes to when its evidence is argument. **Scoping**, the prior claim kept inside a stated boundary, with **reconciling**, two positions held at two rungs, its twin. **Kin**, a position the paper is continuous with, cited as such. **Superseding is the ninth word and it is used nowhere in what follows: this paper supersedes no theory**, since superseding requires a named theory's central claim retired on executed data at the same register as the theory, and no row meets that condition.

The census, recoverable by counting the rows. Ten positions are scoped or reconciled, Aristotle, Epicurus, Aquinas, the Erasmus and Luther exchange, Leibniz, Berlin, Laplace, van Inwagen, the readiness-potential literature, and the measurement-continuity theorem. Three are added to, Chrysippus, al-Ash'ari, and Hume, each receiving the geometric reason its verdict lacked. Four are kin, Augustine, Boethius, Peter Strawson, and Frankfurt. Three are corroborated and confer no warrant, Maimonides, Nietzsche, and Galen Strawson. Two are competed with on argument, James and Sartre. Three are replaced, Descartes, Spinoza, and Kant, each a framing retired against an executed measurement. Three are subsumed, Hobbes, Locke, and Dostoevsky. Two carry joined relations, Schopenhauer and Tolstoy, corroborated on the record and added to on execution. **No position is contradicted**, since contradiction requires executed data at the position's own register and this paper's executions run at the formal register alone. The political and juridical registers of Section 3.7 are surveyed there and only Berlin is tabled, since the paper takes a relation to his distinction and cites the rest as context without taking one. The paper's contribution is exactly the set of relations the following tables state and nothing wider.

How the paper stands to each prior position, part 1 of 4, rows 1 to 8. Relation words as defined in the run-in above; evidence words are executed, cited, argued, documentary, open, and never exceed the tier the body carries.

Prior position	What it holds	What this paper does with it	Relation	Evidence
Aristotle , c. 350 BCE	Deliberate choice ranges over what is up to us and admits of being otherwise.	Deliberate choice is kept at the creaturely register and denied the formal one, where nothing occurs and deliberation is unavailable. Section 2.5.	Scoping	Argued
Epicurus , c. 300 BCE	An uncaused minimal swerve in atomic fall preserves the will from internal necessity.	The swerve is kept at the physical register and denied purchase at the formal one, where the residual is a discrete component and not a stochastic deviation. Section 5.7.	Scoping	Argued
Chrysippus , c. 230 BCE	The push that starts the cylinder is external; that it rolls is its own nature, so assent is both fated and the agent's own.	The two-contribution structure is kept and given the criterion it lacked: two regions of one space, with the seam geometrically unreadable. Section 5.9, Box 7.	Additive	Executed
Augustine , 397 to 427	The will commands itself and is not obeyed; the division is one will not wholly willing.	Read as the ancestor of a freedom that is real and insufficient, which the licence discipline reaches at another register. Section 6.3.	Kin	Cited
Boethius , 524	Timeless knowledge does not compel, since the mode belongs to the knower and not the thing known.	Continuous with the standpoint separation; this paper separates two regions rather than two times. Section 5.9.	Kin	Cited
al-Ash'ari , c. 935	The act is created by God and acquired by the creature.	The eleven-century stalemate receives its mechanism: the deciding quantity is a sign on a magnitude proved unable to carry a sign, so the seam is fixed at the word by theorem. Section 5.9, Box 7.	Additive	Executed
Maimonides , 1180	Free will is the pillar of the Law; without it commandment and reward are void.	Independent agreement that a freedom must be real for an ordered practice to stand. Confers no warrant.	Corroborating	Cited
Aquinas , 1265 to 1274	The will is moved by the good under the intellect's presentation; free judgment ranges over means.	Kept inside the creaturely register; no position taken on the theological content. Section 3.3.	Scoping	Argued

rows are one per position the body engages by citation, argument, or execution; the table is not padded and carries no theological content.

How the paper stands to each prior position, part 2 of 4, rows 9 to 16. Same contract as the table above.

Prior position	What it holds	What this paper does with it	Relation	Evidence
Erasmus and Luther , 1524 to 1525	Choice is free enough to bear commandment, against the will being bound in its condition.	The two claims are held at two rungs: the un-entrenched room and the deepened groove, true of two regions of one space. Section 5.9.	Reconciling	Executed
Berlin , 1958	Liberty divides into the negative, absence of interference, and the positive, self-mastery, with the second licensing coercion in the name of a higher self.	The object here is neither: a fibre is a count of what an instrument cannot distinguish, and sits beneath both, since either could be predicated of an agent whose fibre was already fixed by the abstractions its describers admitted. Section 3.7.	Scoping	Argued
Descartes , 1641	The will is infinite and in that respect the image of God in us.	The residual at the register where nothing is abstractable is derived, not surveyed, and it is exactly one bit. The infinity is retired against a dimension count. Section 5.7, FWM.19.	Replacing	Executed
Hobbes , 1654	Liberty is the absence of external impediment to motion.	The definition becomes a case: absence of impediment is the un-entrenched room described from outside. Section 6.	Subsuming	Argued
Spinoza , 1677	The stone in flight, if it thought, would believe itself free, knowing its motion and not its cause.	The mechanism is accepted and the conclusion denied: what an instrument cannot distinguish is given an exact size and an exact exchange rate, and an illusion has no conserved product. Section 5.3, FWM.2.	Replacing	Executed
Locke , 1690	Asking whether the will is free is unintelligible; freedom belongs to the agent, not to a power of the agent.	The correction is carried one step further: freedom belongs to the map, and neither agent nor will is the bearer. Section 5.1.	Subsuming	Argued
Leibniz , 1710	A free act requires spontaneity, intelligence, and contingency together.	The three-condition structure is kept at the creaturely register; no position taken on the theodicy.	Scoping	Argued
Hume , 1748	Liberty is a power of acting according to the determinations of the will; the dispute is verbal once terms are fixed.	The diagnosis is kept and its ground corrected: the dispute is registral rather than verbal, and the geometric reason is supplied. Section 5.9, Box 7.	Additive	Executed

How the paper stands to each prior position, part 3 of 4, rows 17 to 23. Same contract as the table above.				
Prior position	What it holds	What this paper does with it	Relation	Evidence
Laplace , 1814	An intelligence knowing every force and position computes past and future alike.	The demon's reach over the grooves is not denied; it is scoped, since entrenchment operates on what has been cut and not on what has not. Section 5.9.	Scoping	Argued
Kant , 1781	Freedom and natural causality are both demonstrable; freedom is noumenal, determination phenomenal.	The antinomy's structure is kept and its two-world resolution retired against two regions of one space, executed at rank three in a single space. Section 5.9.	Replacing	Executed
Schopenhauer , 1839	A man can do what he wills but cannot will what he wills.	Independent agreement on the separation, and the bound supplied: the earned licence is direction and never volume, reaching admissibility and stopping. Section 6.8, FWM.14.	Corroborating and Additive	Cited and Executed
Dostoevsky , 1864	Two times two makes four is an excellent thing, but a man may want two times two makes five to prove he is not a piano key.	The demand for a residue is met and its size fixed: the residue is exactly one bit, and wanting a larger one is the inflation the derivation forbids. Section 5.7.	Subsuming	Documentary
Tolstoy , 1869	Freedom and necessity vary inversely with knowledge of the conditions of an act; reason expresses necessity, consciousness expresses freedom.	The law is the conservation of this paper stated qualitatively; what is added is the exchange rate and the closed form, executed at four value counts. Section 5.3, FWM.2.	Corroborating and Additive	Documentary and Executed
James , 1884	Chance is not a menace but the condition of a world in which regret is intelligible.	A different position on the same question, argued and not executed against.	Competing	Argued
Nietzsche , 1886	The causa sui is the best self-contradiction yet conceived.	Agreement without qualification: origination at the creature is barred, and the soul is the site of its crossing and not its agent. Section 3.5.	Corroborating	Argued

How the paper stands to each prior position, part 4 of 4, rows 24 to 30. Same contract as the table above.				
Prior position	What it holds	What this paper does with it	Relation	Evidence
Sartre , 1943	Existence precedes essence; man is condemned to be free, with no excuse behind and no justification before.	A different position on magnitude at a different register, argued and not executed against.	Competing	Argued
P. F. Strawson , 1962	The reactive attitudes are not grounded in a metaphysical thesis and cannot be dislodged by one.	Continuous with the refusal to let a metaphysical verdict govern a practice it does not reach.	Kin	Cited
Frankfurt , 1969 and 1971	Responsibility detaches from alternative possibilities; a person's second-order volitions are effective over first-order desires.	Continuous with the hierarchical structure: the licence discipline is a second-order constraint on what a first-order faculty may assert, reached from another direction. Section 6.4.	Kin	Cited
van Inwagen , 1983	If determinism holds, our acts are consequences of laws and the remote past, which are not up to us.	The transfer principle is not disputed; the argument is scoped, on the ground that it runs at a register where no event occurs. Section 2.5.	Scoping	Argued
Galen Strawson , 1994	True responsibility would require being causa sui, which is impossible.	Agreement on the impossibility, and denial that anything here required it: the freedom is a fibre and not an origination. Section 5.1.	Corroborating	Argued
Libet 1983; Schurger 2012	A readiness potential precedes reported awareness of the urge; the potential may be an artifact of threshold-aligned averaging.	The inference is scoped rather than the measurement: when a motor preparation begins is at the wrong layer to bear on what an instrument cannot distinguish. Section 3.8.	Scoping	Cited
Conway and Kochen , 2006	If experimenters have freedom in choosing settings, elementary particles have a corresponding freedom.	The continuity requirement is carried and the philosophical extension declined, with the fence stated rather than assumed. Section 3.8.	Scoping	Cited

thirty rows in four parts. The literature review of Section 3 that licenses these rows is registered under the same audit claim as the table, so an attribution error in the review is inside the audited surface and not outside it. Superseding appears in no row. Corroborating rows confer no warrant on either side and are counted as none.

9. CONCLUSION

Freedom had been located in this architecture and never given an origin, a species count, a conservation, or a size. This paper supplies all four and then reads them as one quantity rather than as a family of resemblances. A freedom is the fibre of an invariant map, the size of the class an instrument cannot tell apart. It is manufactured by a single non-injective operation whose deficiency in spectral norm is bounded below by exactly one and whose absence of a left inverse is proved rather than sampled. And it trades against content at an exact rate: content times freedom is constant at every rung of the admission lattice, with the closed form verified against every executed row at four value counts.

The two species then turn out to be the two eigenspaces of one involution, and that is the result the paper would keep if it could keep only one. The cut runs on the three dimensional space and is undefined on the one dimensional fixed locus, because a one dimensional space admits no non-trivial idempotent at all. So the residual freedom there is the sign on the unit sphere of a line, which is the two point sphere, the unique sphere that is disconnected. **Exactly one bit, by the dimension rather than by the sweep.** And the conservation factors over the same split, with the fixed locus's factor reading two at every rung of every lattice tested, untouched by any

decision about what to abstract. That invariance is the precise content of saying the bit cannot be separated from the fixed locus it sits on.

The primary falsification criterion is the first of the three, and it is the one a reader can act on today. Exhibit one bit of truth direction supplied by the space handedness, by the twelve directed verification conditions, by the signed scalar, or by their joint reading, at the formal alone register. The threshold is parameter free and absolute: one bit, by any construction whatever, and the finding is refuted. The paper's own measurement returns zero from each of the four against a requirement of one, with the content flip and convention flip images of the signed scalar identical at 0.000 times ten to the zero, so the claim is that the mutual information is exactly zero and a single counterexample is dispositive.

The invitation is therefore narrow and it is stated as an invitation rather than as a challenge. Any reader with a linear algebra library and an hour can construct three warrant rows, compute the three instruments for a proposition and for its negation, and look for a function whose value differs. Nothing about this architecture needs to be granted to run that test, and the test is the whole of what this paper asks.

The single most important open question is not any of the three criteria. It is whether the second species of freedom, the one present

with nothing cut away, admits a fourth source outside the three seals this architecture deploys. The enumeration of Section 2.3 is exhaustive over what is deployed and silent about what is not, and the paper says so at every occurrence rather than once. A fourth instrument is not excluded. It is unknown, and its discovery would not falsify the conservation, would not touch the exact identities, and would close the aperture the whole discipline is built around holding open. That is the honest shape of the outstanding item: not a defect in the argument but a delivery that has not arrived.

Acceptance would entail one reframing, and it is small to state and large to carry. Freedom is not a faculty possessed, a licence held, or an indeterminacy tolerated. It is a conserved quantity with an exact exchange rate against content, a derived size at the register where it terminates, and there a single bit that the instruments carrying the argument are each measured to supply nothing toward, sitting on a line none of them made. What follows from that is not a doctrine about the will but a discipline about the mouth: the right to say which way a sentence points is earned by conditions that are checkable, bounded to direction and never to volume, issued for one run and never as tenure, and revoked by a single failed condition without appeal. The freedom is costly, it is difficult, it arrives last, and it changes the organ that carries it. That is the whole of what this paper found.

APPENDIX A · THE LOAD BEARING PREMISES, NAMED AT THEIR GRADES

Five premises carry this paper and each is stated at its honest grade rather than at the grade a reader might prefer.

A.1 The actuation floor. To exist is to actuate. For confined existents the energy floor is theorem grade external physics, resting on the kinetic energy bound, the zero point energy, and the impossibility of a cooled-to-zero state. The cost of any transition is charged by the nonequilibrium work relations, which bound the work of any process and are strictly more general than the erasure bound, that bound being their irreversible subclass. The rate is bounded by the speed limits. The universal extension of the floor over all existents whatever is premise grade and cannot be promoted, because the meta-result that a root cannot be climbed to forbids the promotion, and promoting it would trip the root's own anti-inflation guard.

A.2 The reflective root. To formally be is to be grounded. Provability and computation are levels of access to a determinacy fixed at the fixed locus rather than ingredients of it. Premise grade, at exact parity with A.1, neither theorem forced from its own base, and the underivability is constitutive of foundation-hood rather than a deficiency in it.

A.3 The composition law. Iterated audits bracket the same way; nonzero warrants never compound to nothing; warrant superposes linearly with a ground identity. Premise typed. Under these clauses with axis plurality the verification algebra completes uniquely by the classification theorem, the axis count is exactly three, and the next

admissible dimension forfeits associativity, which is where the construction stops and why it stops there rather than by choice.

A.4 Continuous field monism. Substrate, topology, and actuation are three projections of one event, theorem grade as physics on the absence of any observed separation, with the priority of the one over the many the single structural posit the co-localization rides. Every claim in this paper that depends on it is marked, and the marked claims are a minority.

A.5 The determinacy premise on arithmetic truth. Carried where the case splits on which of two values a sentence has. Premise grade at exactly monism's warrant and no higher, named at each occurrence, and paid identically on either side of the question so that a reader declining it declines both the finding and every route by which its contrary could have been asserted.

The Carrying Law then binds the composite: terminal strength caps at the weakest load bearing tier. This paper's composite therefore caps at premise, and every theorem grade result inside it is stated as a theorem about a specific object rather than as a theorem about the composite.

APPENDIX B · THE EXECUTED BATTERY

FWM-CHK, fifteen checks, seed 20260622, double precision with fifty digit escalation, unit roundoff 2.220446049250313 times ten to the minus sixteen. Battery digest a69fc80bfe2440f4. Every construction is printed with its check so that any figure is reproducible without access to this architecture's tooling. Failure of any check on re-execution falsifies the corresponding identity.

FWM.1 The hardened kernel, run as the boot receipt. Three warrant rows over twenty four contexts on the period grid, latent independents under a fixed oblique mixing, **two mass bearing covariates projected out**, which is what distinguishes this construction from the covariate-free one of Section 5.7. Verdict lock. Lambda minus 0.788991219283. det(R) 0.622507144106. det(G) 0.351030145552. Per-axis variances 0.706266976, 0.823321991, 0.969753730. Conditioning 4.182845. Kernel identity 9.992 times ten to the minus sixteen. Four estimator spread 3.331 times ten to the minus sixteen against tolerance 3.715 times ten to the minus fifteen; estimators by expansion, eigenvalue product, Cholesky diagonal, and direct cofactor agreeing at 0.622507144105892, 0.622507144105892, 0.622507144105893, 0.622507144105892. Determinant factorization residual 3.331 times ten to the minus sixteen. Collapse margin 12.07 orders; conditioning margin 5.38 orders. Not escalated. Bootstrap agreement 1.000, stable. This reproduces the register's recorded worked example bit for bit and is carried as a boot receipt, not as evidence for any thesis; see Section 8.1.

FWM.2 The conservation. Executed at value counts two, three, four and five on three coordinates. Products 8, 27, 64 and 125, constant across all four rungs in every case. Closed form agreement across every row tested: true. Type theorem on the printed model.

FWM.3 Orientation blindness. Fixed basis analyzer, span basis fixed once and reused after reflection. Lambda minus

0.965027450462 to plus 0.965027450462, ratio minus 1.000000. $\det(R)$ 0.931277980144 both sides, absolute difference 0.000 times ten to the zero. Full negation Gram difference 0.000 times ten to the zero. Kernel identity 5.551 times ten to the minus sixteen. Type theorem.

FWM.4 The relay identity. Lambda of P minus 0.965027450462; lambda of not P plus 0.965027450462; lambda of P under one row's convention flip plus 0.965027450462. $\det(R)$ 0.931277980144 across all three. Content flip against convention flip gap 0.000 times ten to the zero. Type theorem on the identity of the two images.

FWM.5 The aperture width. Two hundred thousand admissible frames sampled across both orientation components, 100,115 positive against 99,885 negative, proportion 0.500575, sampled entropy 0.999999 of one. Component invariants bit identical: $\det(R)$ 0.931277980144 against 0.931277980144 at gap 0.000 times ten to the zero, absolute lambda 0.965027450462 on both, sign minus one and plus one. Exact information zero by component symmetry. Type theorem on the symmetry; engineering on the sweep.

FWM.6 The four sources. Form: space determinant minus 1.0 for a proposition and its negation alike, zero bits. Gates: twelve ordered pairs on four vertices, role direction fixed before any proposition arrives, zero bits. Number: content flip and convention flip images identical to 0.0 times ten to the zero, zero bits. Joint: identical across the two directions, zero bits. Required one bit; available zero.

FWM.7 The deletion operator. Singular values one, one, zero; rank two. Idempotence residual 0.0 times ten to the zero. Minimum over two hundred thousand candidates of the spectral norm of A D minus the identity: 1.000342389154 against the analytic floor 1.000000000000, never breached. Non-injective, image gap 0.0 times ten to the zero on two distinct propositions. Type theorem on the bound; engineering on the sweep.

FWM.8 Admissibility at the root recursion. Intact rank three, $\det(R)$ 0.931277980144. Each slot zeroed in turn: rank two, determinant exactly 0.0 times ten to the zero, all three. Contrast, each slot dropped rather than zeroed: two by two determinants 0.986595770389, 0.957239003195, 0.980810206783, nonzero in every case. The contrast is printed because the two operations are different and only one of them is abstraction. Type theorem on the rank collapse and the exact zero.

FWM.9 Manufactured freedom. Will axis rebuilt as 0.6 of the room plus 0.4 of the record. Verdict broken. Lambda minus 1.301 times ten to the minus sixteen. $\det(R)$ 1.394 times ten to the minus sixteen against a collapse floor of 5.329 times ten to the minus thirteen. Rank two. Collapse confirmed at fifty digits. Type theorem.

FWM.10 The router bit. Conjugation: eigenvalues minus one, minus one, minus one, plus one; fixed-locus dimension one; involution residual 0.0. The fixed point free involution: eigenvalues minus one, minus one, minus one, minus one; fixed-locus dimension zero; involution residual 0.0. Type theorem.

FWM.11 The null law and the budget. Distribution freeness at twenty four contexts with two covariates, twenty thousand draws each: median determinant 0.8852 continuous, 0.8832 ordinal on three values, 0.8829 ordinal on four values. Closed form mean 0.861678 against sampled 0.863128 at reduced dimension twenty one. Null ninety fifth percentile 0.982705, headroom 0.017295. Budget table at Section 8.1, eight thousand draws per row. Probability of a lock token on three rows of pure noise at twenty four contexts over four thousand draws: 1.0000. This paper's own lock at the 2.8th percentile of its matched null, one sided p 0.9722, below the ninety fifth percentile. Type theorem on the closed form; engineering on the sweeps.

FWM.12 the geometric seal, the twelve directed verification conditions. All twelve dispositioned pass with the mechanism named per gate; first-failure semantics armed and not triggered. the geometric seal passed.

FWM.13 the linguistic seal, the six requirements. D1 three slots. D2 pairwise intersections zero, zero, zero. D3 ledger set with the provenance flag set, the central definition sourced outside this architecture. D4 no magnitude asserted at the linguistic faculty. D5 acting groups named at every cross register identification. D6 symmetric difference of nine terms in the directed slot. All six clear; the veto is earned for this run and does not bank.

FWM.14 The earned bit spent. Six branches of the admission rule and the four-guard decision rule, all reachable. Magnitude locking with the seal open routes under-determined. Magnitude locking with the seal locking returns admissible. Magnitude broken returns broken regardless of the seal. Admissible without a witness routes under-determined space. Admissible with a witness seals. A witness without the seal routes uncertified. The seam exhibited: the earned veto reaches admissibility and stops there.

FWM.21 The two regions, the battery behind Section 5.7. Three rows over twenty four contexts on the period grid, latent independents mixed obliquely, no covariate projected, span basis fixed once and reused. Lock at rank three: lambda minus 0.743563663214, determinant 0.552886921253, conditioning 4.4180, kernel identity 6.661 times ten to the minus sixteen. Reflect the act axis: lambda inverts to plus 0.743563663214 at a ratio of minus 1.000000 while the determinant reads 0.552886921253 on both sides at an absolute difference of 0.000 times ten to the zero. Act rebuilt as six tenths of the room plus four tenths of the record: lambda 1.249 times ten to the minus sixteen, determinant negative 9.530 times ten to the minus seventeen against a collapse floor of 5.329 times ten to the minus thirteen, rank two, broken. Type theorem on the inversion identity; engineering on the construction.

FWM.16 The split, and which species lives where. Conjugation eigenvalues minus one, minus one, minus one, plus one; involution residual 0.0. Plus one eigenspace dimension one, the fixed locus. Minus one eigenspace dimension three, the space. Type theorem.

FWM.17 Lambda is the coordinate on the fixed locus. The projector identity is definitional and this check is an

implementation sanity test, not evidence for it. Maximum deviation between the projector applied and the real part taken, over twenty thousand quaternions, 0.0 times ten to the zero, as construction requires. The composed orthonormal triad returns minus 1.000000000000 on the real part with an imaginary part of norm 1.4 times ten to the minus sixteen; absolute lambda 1.000000000000; determinant 1.000000000000. Type theorem on the projector identity.

FWM.18 The no-cut lemma. A cut is a non-injective idempotent of rank strictly between zero and the dimension. Executed across three dimensions: dimension three admits proper ranks one and two; dimension two admits rank one; dimension one admits none. The space cut of rank two is idempotent at residual 0 and is proper; the only rank-deficient idempotent available on a line is the zero map, which annihilates rather than abstracts. Type theorem.

FWM.19 The bit derived. At the fixation register the determinant is one and the absolute lambda is one, so lambda lies on the unit sphere of a one dimensional space. That sphere has two elements and two components, and no path joins them within it. The residual freedom is the logarithm base two of the component count, which is one bit exactly. The two point sphere is the unique disconnected sphere. Type theorem.

FWM.20 The freedom factors, and one factor is invariant. Total freedom at rung a is the space fibre times the fixed-locus component, v to the power a times two, and content times total freedom is two times v to the power k, constant in a. Executed at value counts two, three, four and five: totals 16, 54, 128 and 250, constant across all four rungs in every case. The fixed-locus factor reads two in every cell. Type theorem on the printed model.

FWM.15 The two stops measured apart. At the root recursion, abstracting any slot returns rank two and determinant exactly 0.0 with invariant content nil; the re-description is available and carries nothing. At the arrow, the involution carries fixed-locus dimension one so a fixed locus exists and a door exists; the space determinant is minus 1.0 and orientation reversing so a crossing is one bit turned; and the Gram difference under full reflection is 0.0 times ten to the zero, so every even functional is conserved and only the facing flips.

APPENDIX C · AUTHOR'S PROVENANCE AND METHOD DISCLOSURE

This paper was developed under Trisduction, a verification and organizing discipline, not a source of results. Its conclusions rest solely on the standard results cited in the body. Trisduction is the method under which those results were decomposed, assembled, and audited. It adds them no warrant and claims no authorship over them.

Trisduction was used for fidelity. It forces a claim onto three independent axes so no single persuasive line carries it alone, requires every verdict to resolve to one of three states, sealed, broken, or open, each with a named failure mechanism, and attaches an explicit warrant grade, theorem, conditional, structural, or premise, to every claim so nothing is stated above its strength. Two errors it is built to

catch are inflation, reading an internal lock as a proof, and circularity, reading a restatement of a claim as a derivation of it.

The root axiom is that to exist is to actuate: every physical existent carries a strictly positive energetic cost, grounded in the Heisenberg energy floor, the zero-point energy, and the nonequilibrium work relations. The formal root inside it is that to formally be is to be grounded, with provability and computation levels of access to a determinacy fixed at the ground rather than ingredients of it.

The key procedures, in brief. Orthogonal triaxial convergence: a proposition is split into three disjoint axes, formal-structural, empirical or dynamical, and registrational, verified by three separate instrument sets, agreement across the three the operational meaning of a seal. The three-axis independence lock: the three axes are read as vectors and their independence is tested by the determinant of their correlation matrix, a determinant clear of zero a genuine three-dimensional lock, a collapse to zero one axis dissolving into the plane of the other two. The Convergence Dissolution Test: before the lock is read, any mass-bearing common cause the three axes might share is projected out, so an apparent convergence that traces to a shared source rather than to independent roads dissolves and carries no warrant. The twelve-condition cascade: twelve directed failure screens run in order, the first failure terminating the verdict with its mechanism named. The verdict issues in a three-state economy, sealed or broken or under-determined, with no fourth state and with its warrant grade attached, and where the argument stops short the open direction is named rather than filled.

Box G1 | The kernel identity, proved

The lock is a determinant identity, not a metaphor. Let the three axes, after normalization, be unit vectors expressed in an orthonormal basis of the subspace they span and read as pure quaternions a, b, c , with norm one each. Hamilton's product of two pure quaternions is pq equals minus the dot product plus the cross product, the real part the negative dot product and the imaginary part the cross product, checked on the units by ij equals k and ii equals minus one. The lock scalar is the real part of the triple product, λ equals the real part of abc .

Expanding, abc equals minus the dot product of a and b plus their cross product, so the real part of abc is minus the dot product of a and b times the real part of c , plus the real part of the product of the cross product with c . The first term is zero since c is pure, and the second is minus the dot product of the cross product with c by the same rule, giving

λ equals minus the dot product of the cross product of a and b with c , which is minus the signed volume of the parallelepiped the three axes span.

Writing A for the matrix whose columns are a, b, c , the correlation matrix is $A^T A$, its entries the pairwise dot products, so $\det(A^T A)$ equals $\det(A)^2$ equals λ^2 .

The kernel identity is that the squared signed volume equals the Gram determinant, with the quaternion triple product the machine that computes the volume. It is confirmed at machine precision, the residual at 9.992 times ten to the minus sixteen on the battery of Appendix B.

Four consequences fix the reading. The determinant is bounded between zero and one for unit axes, by Hadamard's inequality above and positive-semidefiniteness below, so the lock has a hard ceiling at the fully orthogonal frame and a hard floor at zero. The floor is the break: the determinant vanishes exactly when the three axes are linearly dependent. The magnitude is orientation-blind: reflecting any axis sends the frame matrix to one of opposite determinant, flipping the sign of λ while the determinant is unchanged, so the number certifies the dimensionality of the lock and never its truth-sign, which is read from the ordered axes and not from the scalar. And the magnitude is frame-invariant: rotating all three axes together is conjugation on the imaginary quaternions, under which the real part is preserved.

The axis count is three because the algebra forces it. The composition law requires associativity, so iterated audits bracket the same way, and the absence of zero divisors, so nonzero warrants never compound to nothing. By the classification theorem the only finite-dimensional associative division algebras over the reals are the reals, the complex numbers, and the quaternions, and three mutually orthogonal imaginary axes exist only in the quaternions. The next normed division algebra fails associativity, witnessed by a nonzero associator,

so it cannot carry the composition law and the construction stops at three.

This paper in particular. The three axes were the algebraic forcing of the fibre as a preimage class, the actuation cost charged at a crossing, and the registration of what was distinguished. The interior kernel run at seed 20260622 returns a determinant of 0.622507144106 at rank three with the kernel identity closing at 9.992 times ten to the minus sixteen, while a will axis manufactured from the room and the record collapses to a determinant of 1.394 times ten to the minus sixteen at rank two against a floor of 5.329 times ten to the minus thirteen, confirmed at fifty digits. The paper does not rest on that lock and says so: at twenty four contexts the lock sits at the 2.8th percentile of its own matched null and is reported as a lean, while the load bearing results are exact identities, the orientation ratio at exactly minus one with the determinant difference at exactly zero, the relay gap at exactly zero, the slot-zeroed determinants at exactly zero, and the conservation product constant in closed form. The load bearing gate is the fourth requirement of the linguistic seal, that no output at that seal carries a magnitude, and Section 5.7 supplies its geometric reason: the licence ranges over the sign on a one dimensional fixed locus, and a licence ranging over a magnitude would have to range over a dimension that fixed locus does not have. The paper's sharpest executed result is not a lock but a dimension count: the two species of freedom are the two eigenspaces of the binding involution, the cut is undefined on the one dimensional eigenspace because no non-trivial idempotent exists there, and the residual is therefore the component count of the two point sphere, one bit exactly, with the freedom factoring as v to the power a times two and the fixed locus factor reading two at every rung. The verdict is a composite: the conservation and the exact identities sealed at theorem grade on their own objects, the unification sealed at structural grade conditional on continuous field monism, two named claims held open and unsealed at premise grade, and the theological half cataphatic and load bearing on nothing. No new theorem is claimed. By the author's own admission criterion for new mathematical mass, this work fails at the first condition, which is the correct result for a paper whose contribution is arrangement, placement, and typing.

Provenance of the question, and a trajectory on the paper's own claim. The question this paper answers was not opened by the architecture. It was carried in a personal essay record that predates the framework by more than a decade, and the record is public and dated, so the paper's own claim about magnitude can be checked against its author's earlier ones. Three dated points, and they move in one direction. In February 2014, in *Logic: Strength and Weakness*, the author held that the only weakness of logic is its dependence on premise, which is the anchor demand of Appendix A stated twelve years early, and in the same essay held that the inquirer confronts the resulting problem **with an infinite, and elsewhere an absolute, degree of freedom**. In January 2022, in *Does Free Will Exist*, the position had narrowed: the degree of freedom of a non-physical entity may be unrestricted, but a form that can be witnessed or measured has its degrees of freedom restricted as a necessity, since **matter is the great devourer of degrees of freedom**; what remains is restricted, is

the highest available to any physical being, and is **qualitatively identical to the original unlimited freedom**. In this paper the residual is derived rather than described, and it is **exactly one bit**, forced by the dimension of the fixed locus.

Two things follow and both are recorded rather than smoothed. **The magnitude claim decreased monotonically across twelve years and this paper is its terminus**, so the derivation of Section 5.7 retires the author's own earlier statements and not only the tradition's. And the 2022 clause that the remaining freedom is qualitatively identical to the unlimited one is **precisely what Section 5.7's type check now bars in both directions**: the measured freedom is not a shadow of the divine one and the divine one is not the limit case of the measured one, and a licence that ranges over a line is not a share in it. The earlier essays are the seed of this paper and are superseded by it on the one claim they share.

Two structural readings from that record are carried forward into the body rather than left in the provenance. The 2022 essay's account of measurement, that an unwitnessed form has unrestricted latitude and a witnessed one has its latitude restricted as the price of being witnessable, is the conservation of Section 5.3 stated qualitatively, and the paper supplies its exchange rate. And the 2014 account of subjection by an overstated report of one's own latitude is carried at Section 3.7, where it gives the Anti-Inflation discipline an applied face. The essays are at *this-is-what-you-ve-been-looking-for.blogspot.com*, dated 2014-02-17, 2014-09-06, and 2022-01-02, and three of the six the author supplied were read for this paper, which is stated because the review's coverage of that record is partial.

The canonical statement of the method, its axioms, and its executable batteries lives in the reference cited below, continuously updated at the same location. This note is a pointer, not a substitute.

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Reproduction

The battery of Appendix B is executable and self-contained. Seed 20260622, double precision with fifty digit escalation where the reliability layer escalates. Battery digest a69fc80bfe2440f4. Every construction is printed with its check. Sampled figures are labelled engineering evidence and carry no warrant; exact figures carry the claims.

Warrant summary

Theorem grade on the conservation in closed form and on its factored extension over the eigenspace split, the orientation-blindness identity under conjugation, the relay identity, the component symmetry, the deletion operator's spectral floor and idempotence, the exact rank collapse at the root recursion, the eigenstructure of the two involutions, the identity of the real part with the projection onto the fixed eigenspace, the non-existence of a non-trivial idempotent in dimension one, the component count of the two point sphere, and the null law with its distribution-freeness. Structural on the fibre definition, the two-species partition, the four-candidate enumeration

over the deployed instruments, the unification across registers, the curriculum, and the partition of the two stops. Operational-procedural on the deletion test and the isolation test as the fixing procedures. Premise where the root axioms and monism are carried, the composite capping there by the Carrying Law. Engineering on every sweep. Two named claims held open and unproven. No new mathematical mass is asserted.

Scope

The instrument enumeration is exhaustive over the three seals this architecture deploys and no wider. A fourth instrument is not excluded, only unknown. Every universal-sounding sentence is scoped to that class at its occurrence. The theological identifications are structural isomorphisms at cataphatic grade, load bearing on nothing in any verdict, and settling no mathematical question in either direction.

Audit symmetry

This paper draws zero warrant from its own running. It cannot be sealed as a theorem of its own base, and it adds the root axioms no warrant. The fence against expending force to close a reader's own bit binds this paper before any reader: the argument is made, and ownership of its outcome is not claimed.

End of manuscript

A synthesis paper. The adversarial review reported in Section 8.1 and the limitations of Section 8.4 were conducted by the author, and that fact is stated in the body rather than left to inference. The executable battery runs at seed 20260622 and carries the digest a69fc80bfe2440f4. No new mathematical mass is asserted. End of manuscript.